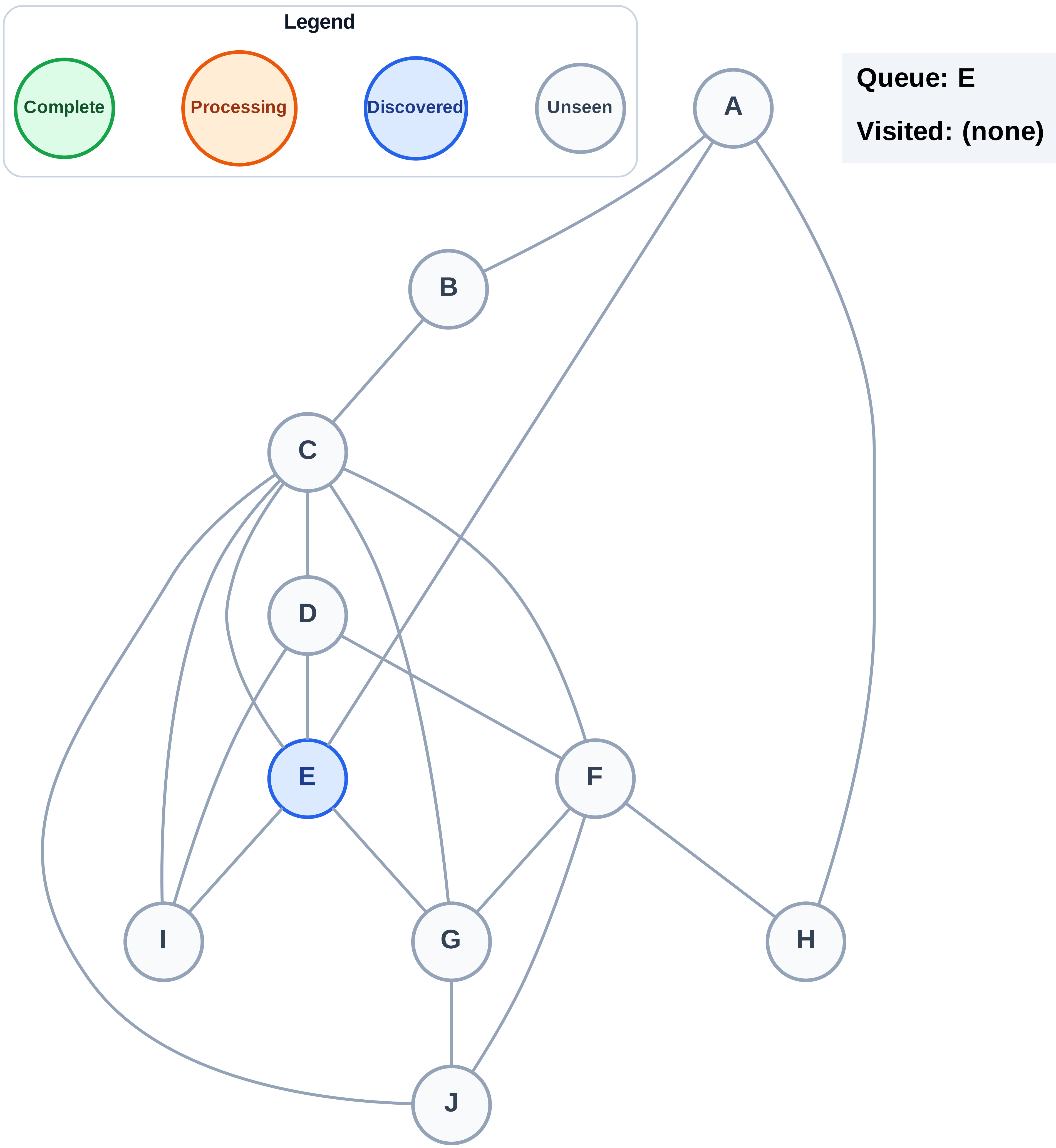
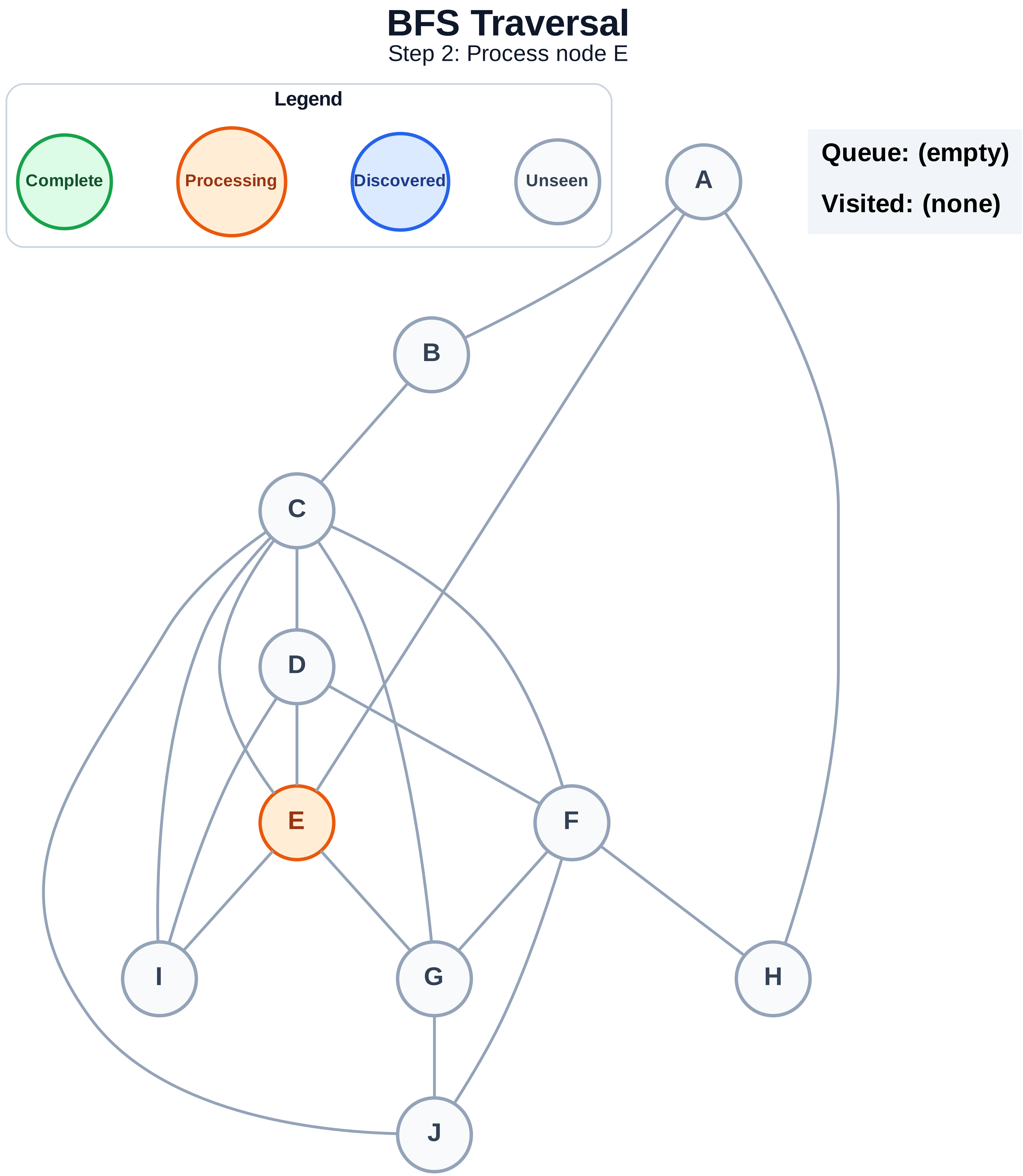
BFS Traversal

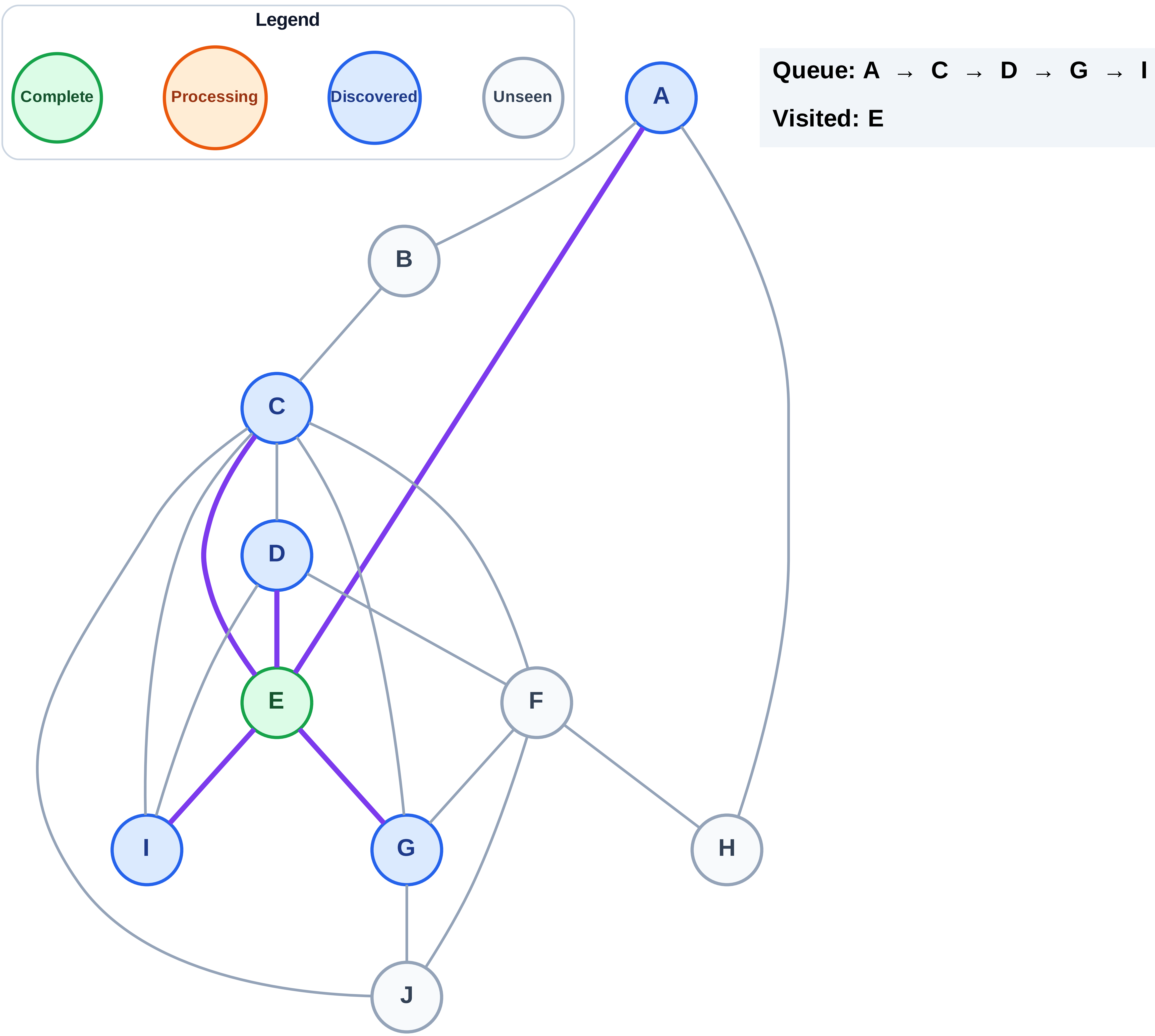
Step 1: Start at E




BFS Traversal
Step 2: Process node E
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: (none)
A
B
C
D
E
F
G
H
I
J

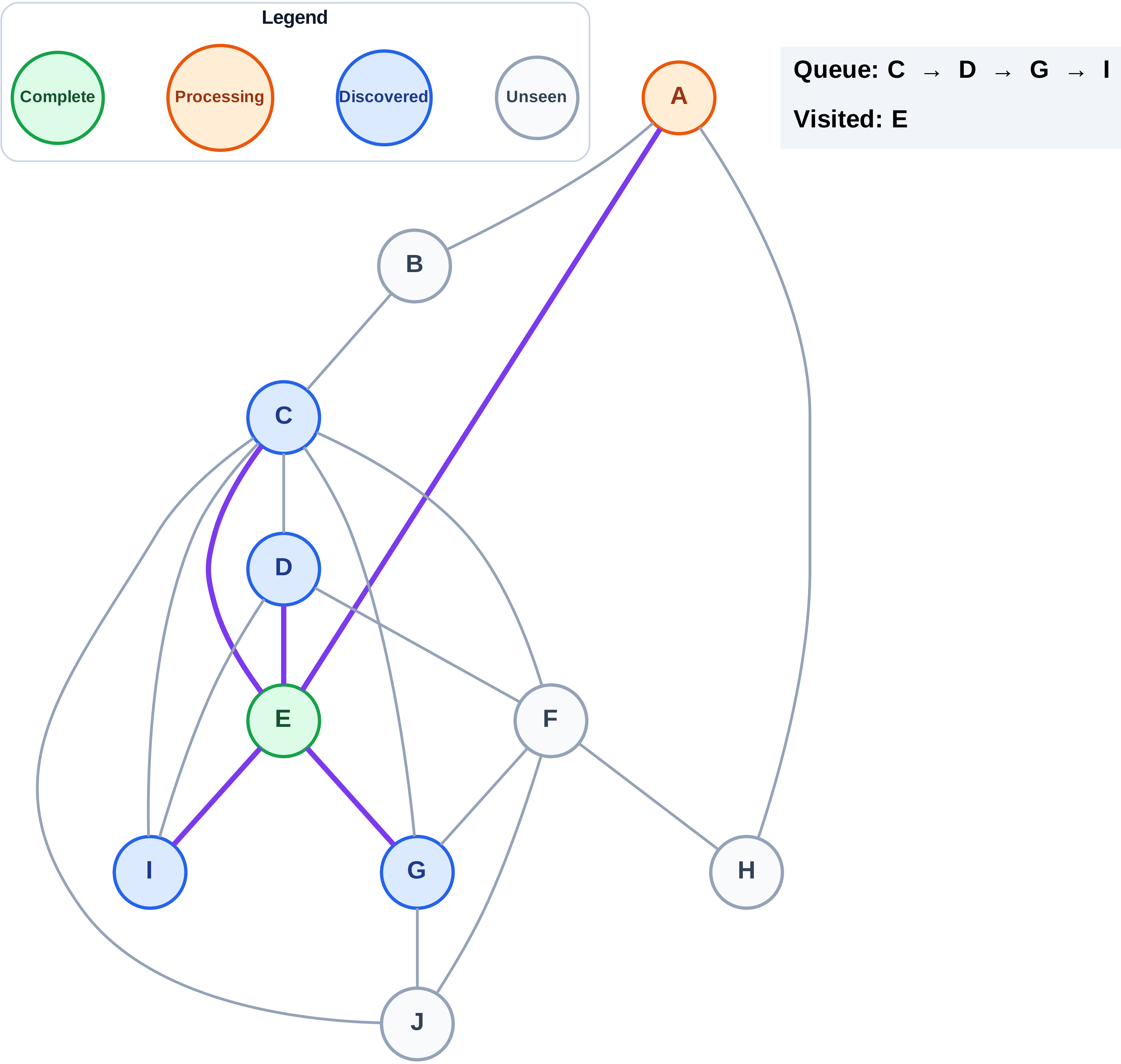
BFS Traversal

Step 3: Discover A, C, D, G, I from E



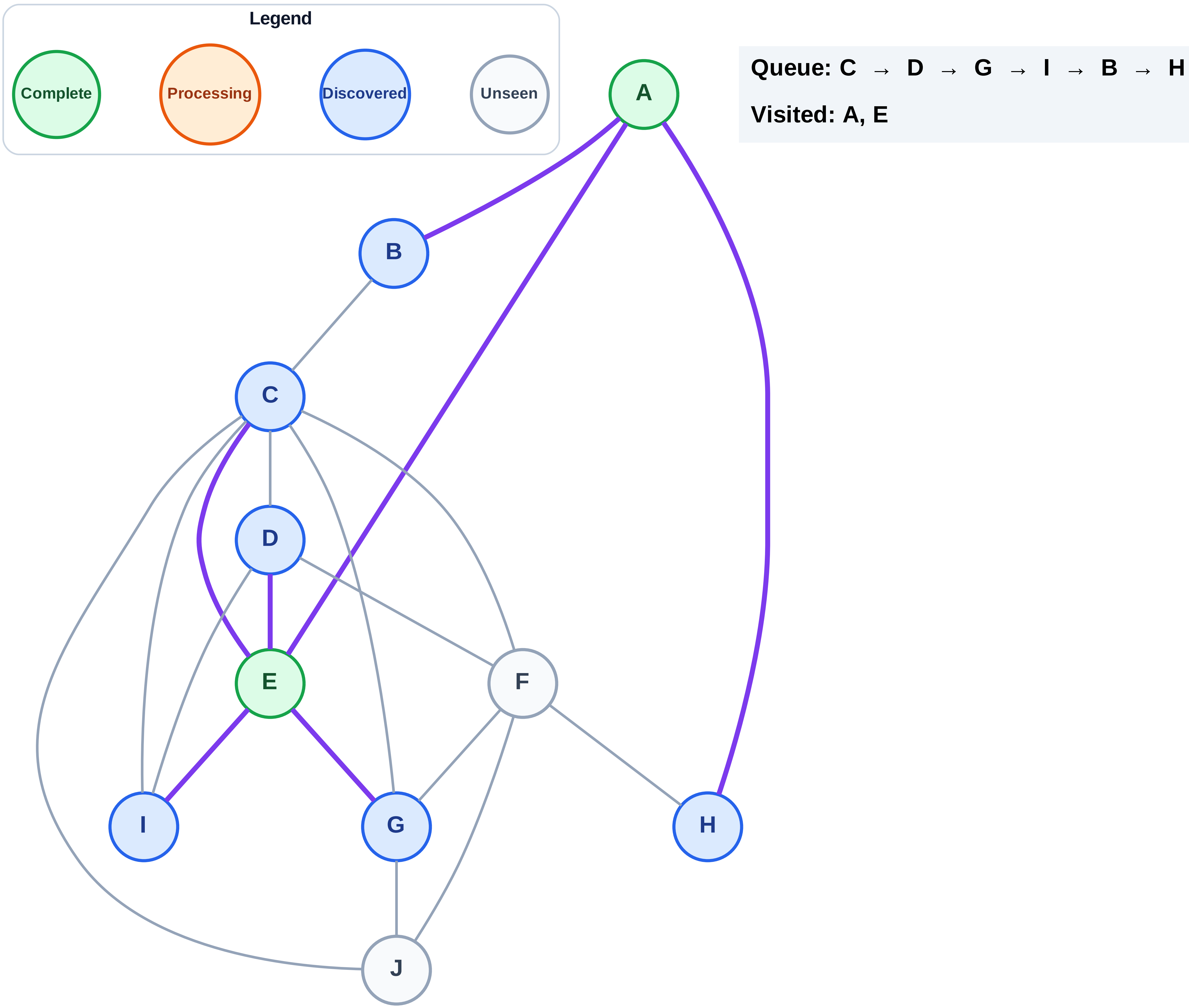
BFS Traversal

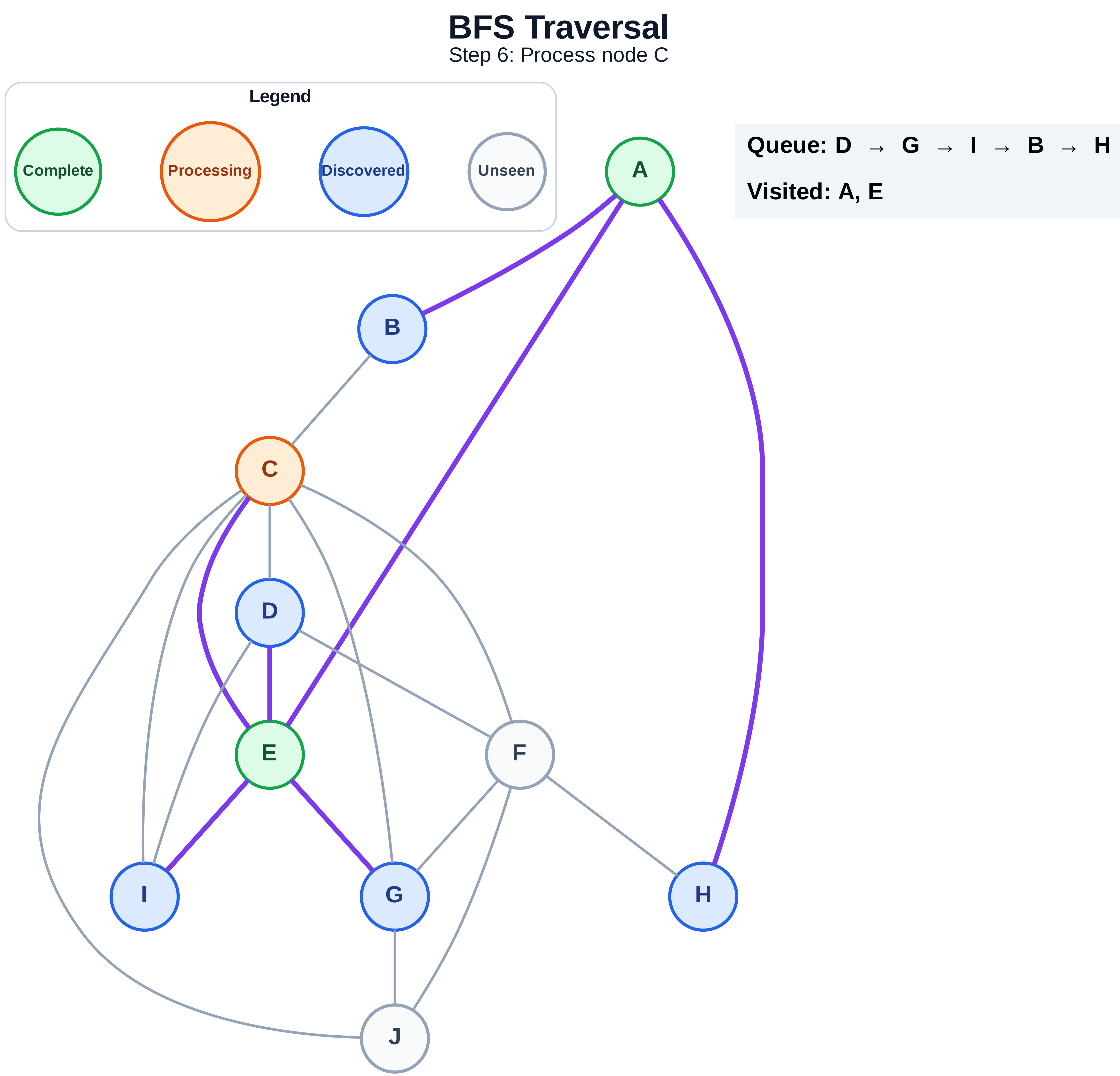
Step 4: Process node A



BFS Traversal

Step 5: Discover B, H from A

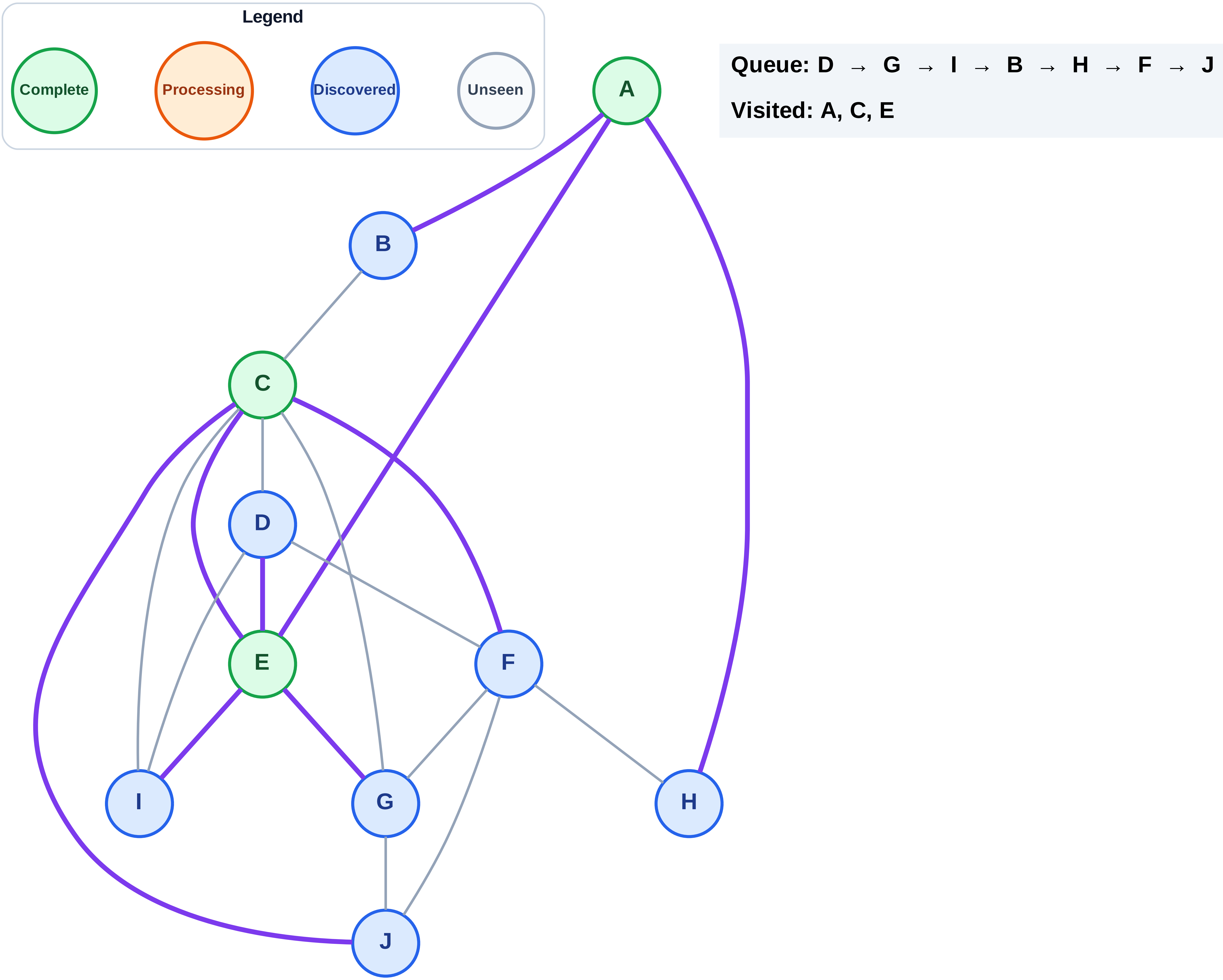




BFS Traversal
Step 6: Process node C
Legend
Complete
Processing
Discovered
Unseen
Queue: D → G → I → B → H
Visited: A, E
A
B
C
D
E
F
G
H
I
J

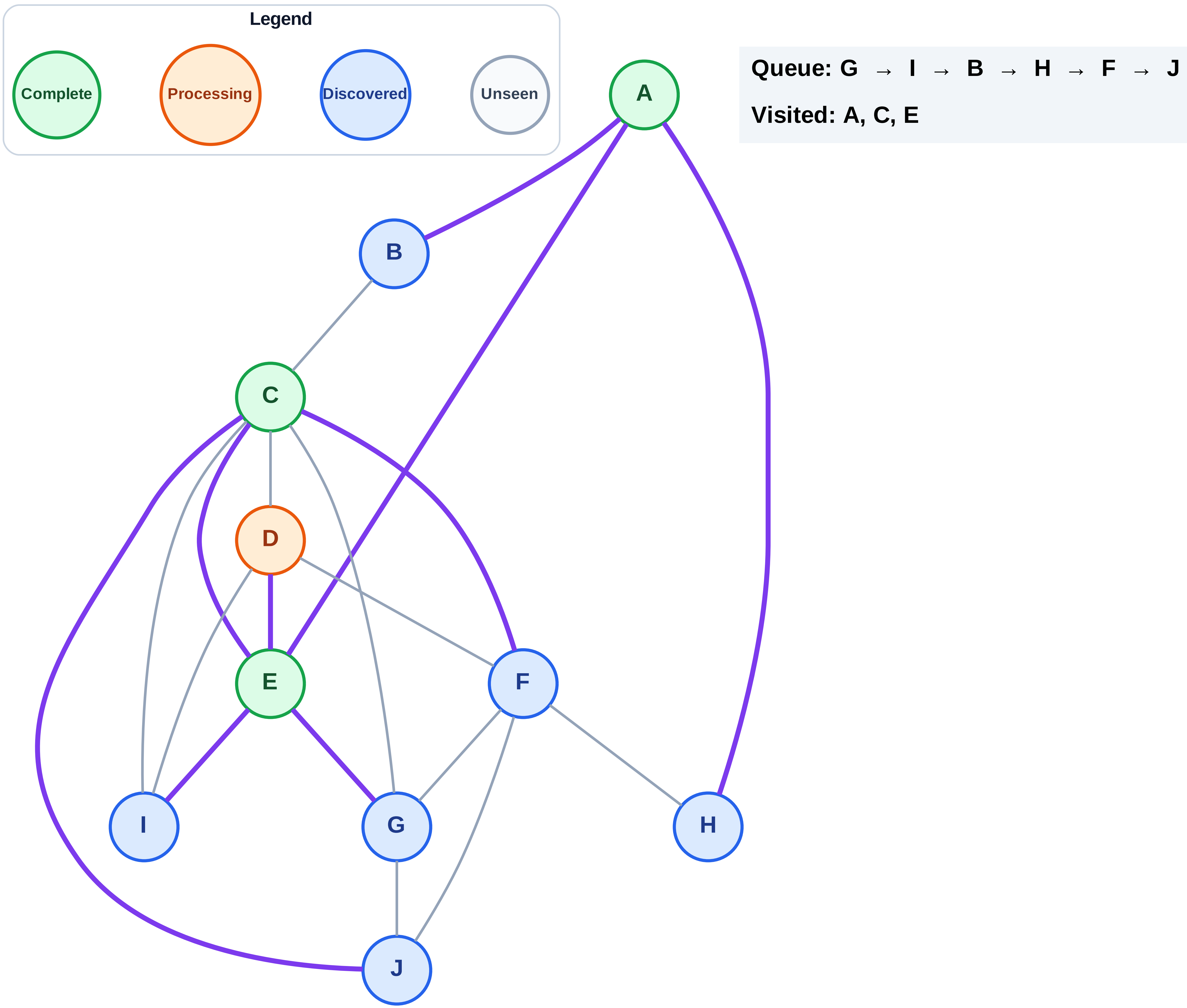
BFS Traversal

Step 7: Discover F, J from C



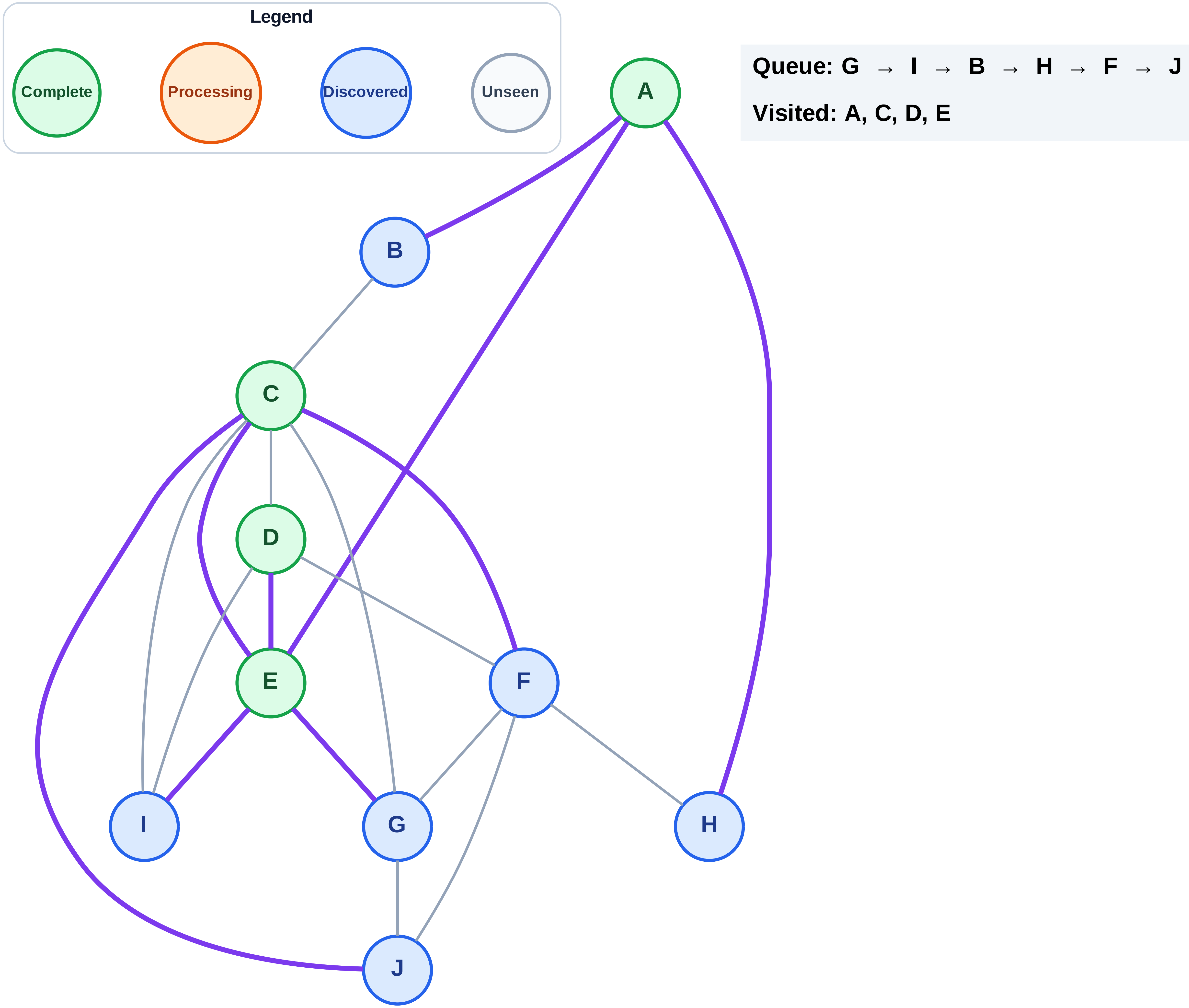
BFS Traversal

Step 8: Process node D



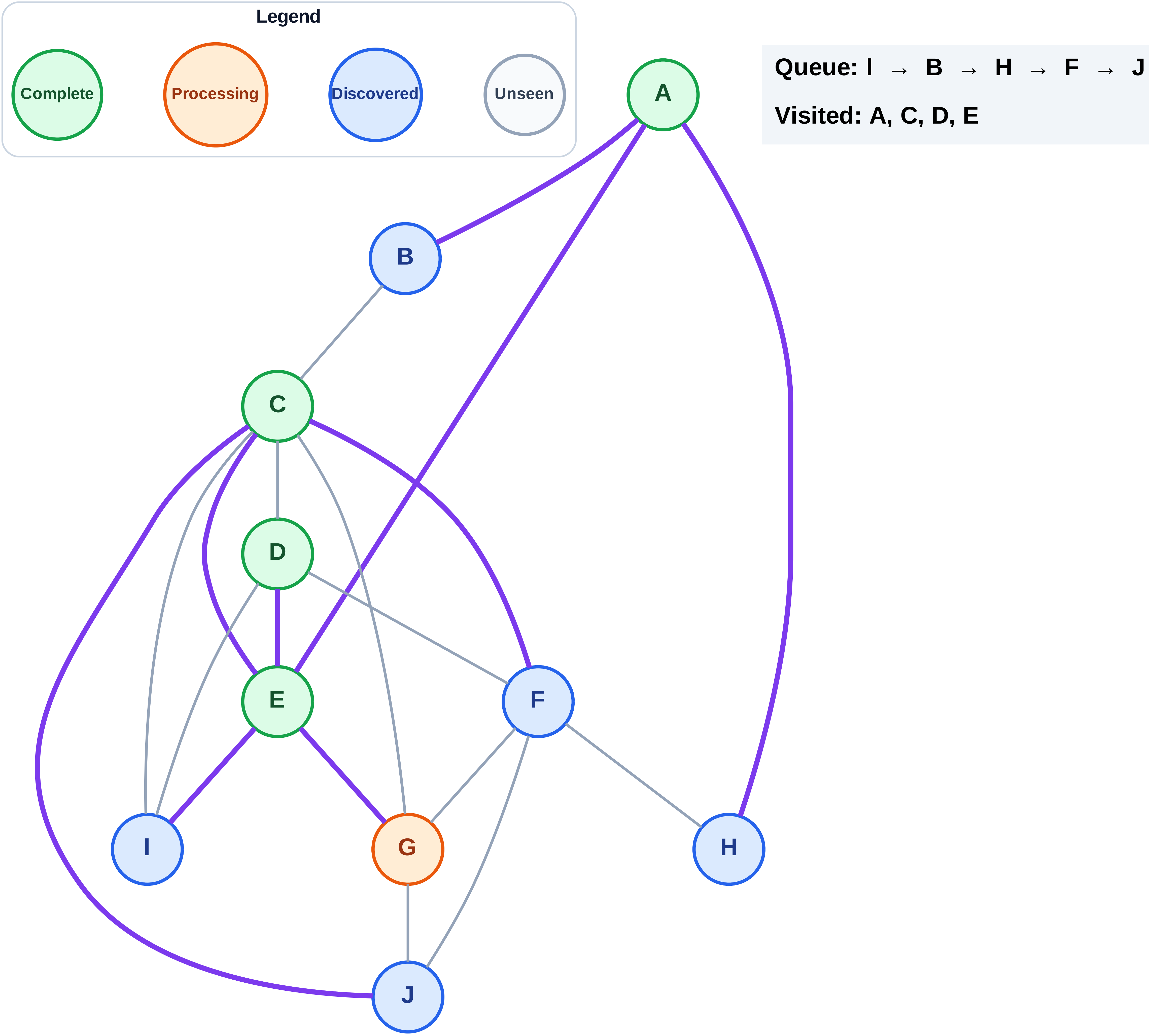
BFS Traversal

Step 9: No new neighbors from D



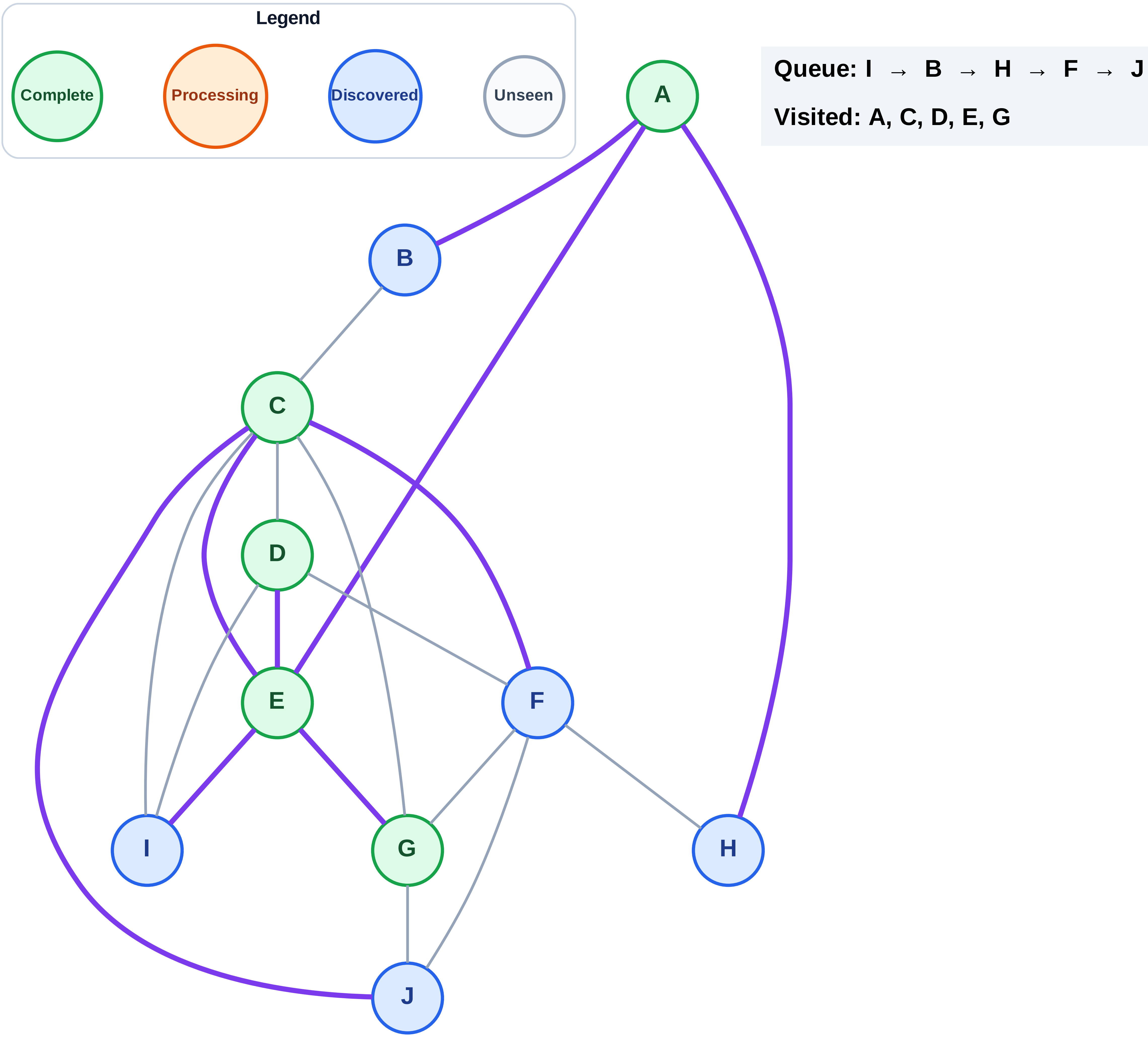
BFS Traversal

Step 10: Process node G



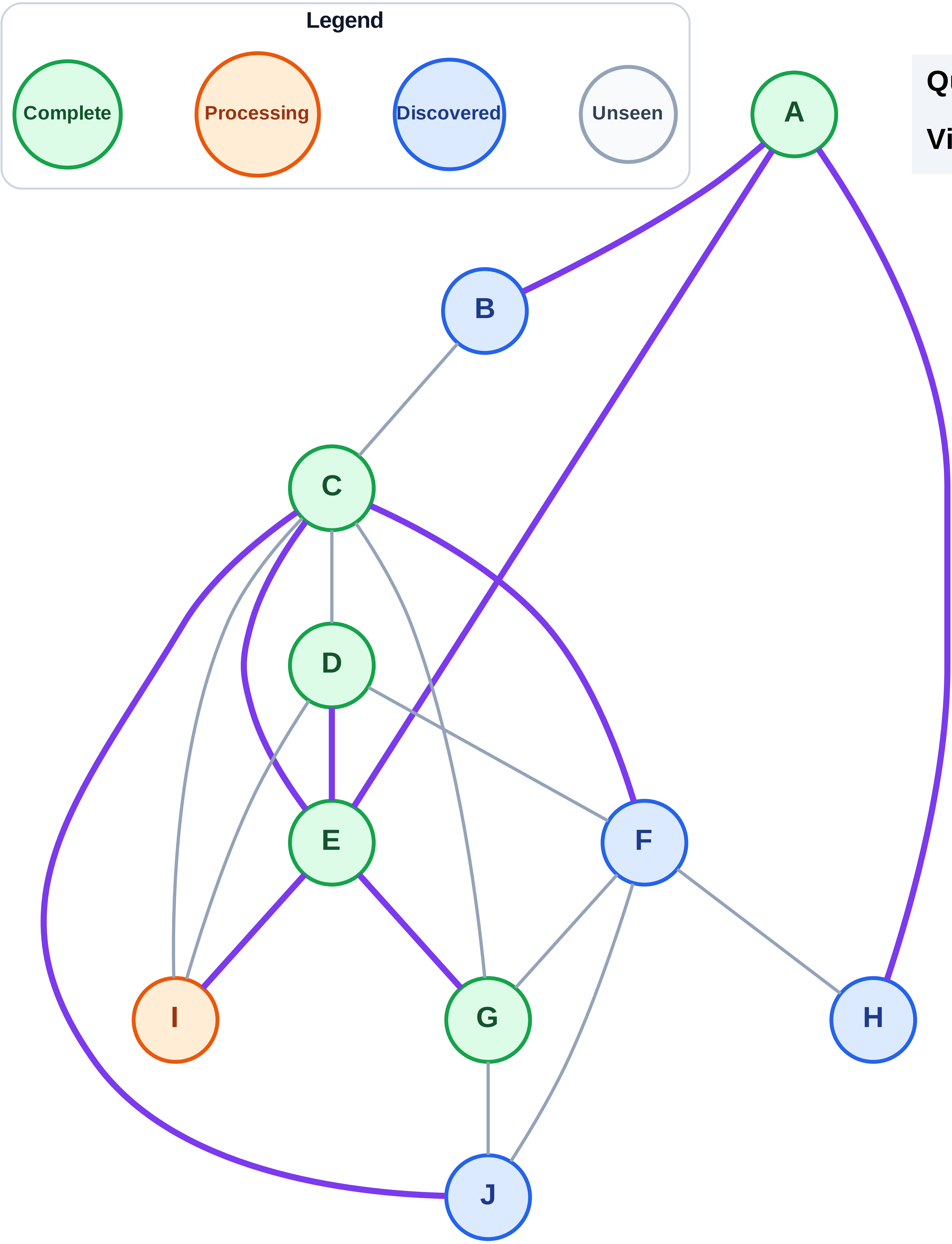
BFS Traversal

Step 11: No new neighbors from G



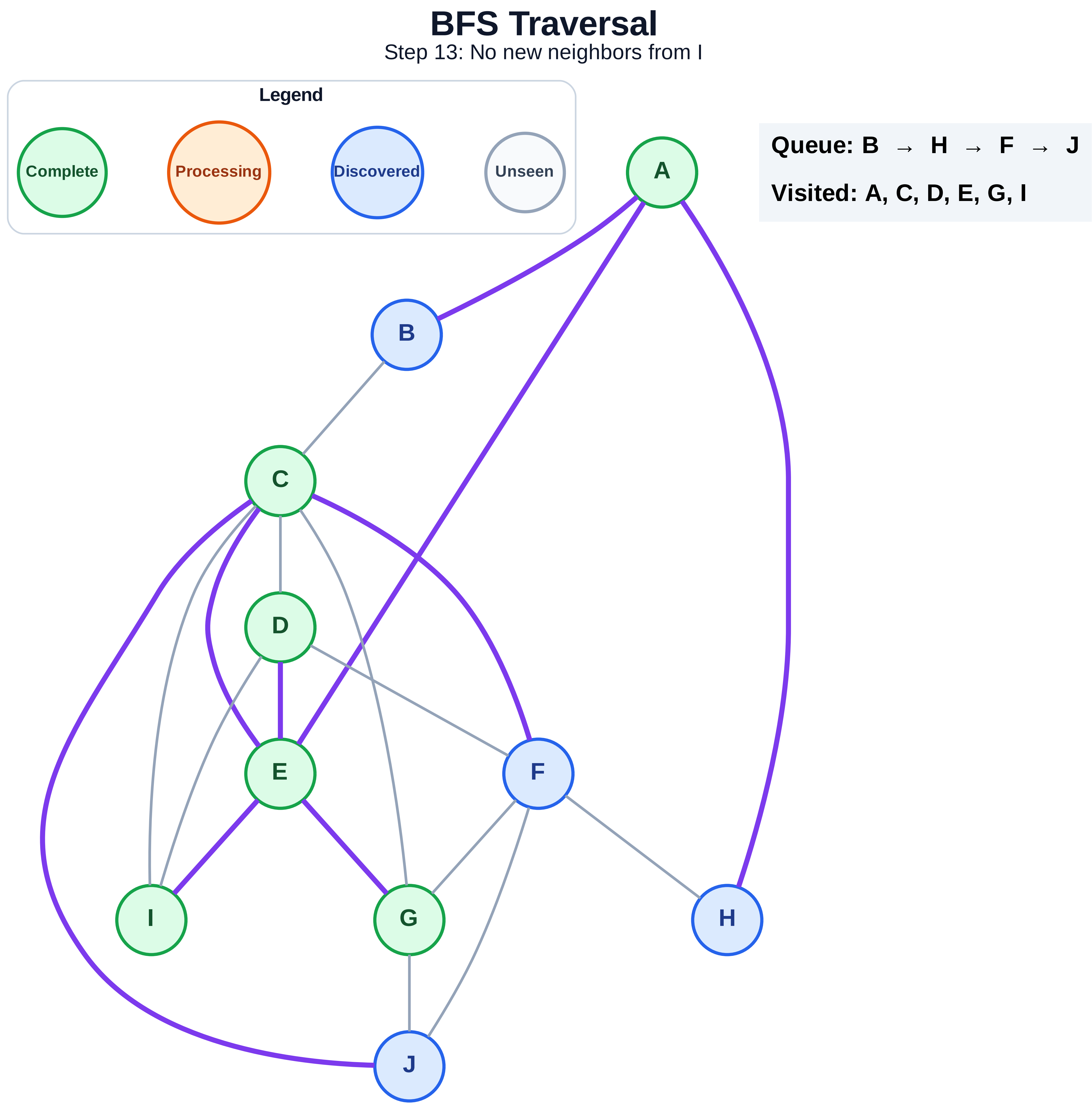
BFS Traversal

Step 12: Process node I



Queue: B → H → F → J

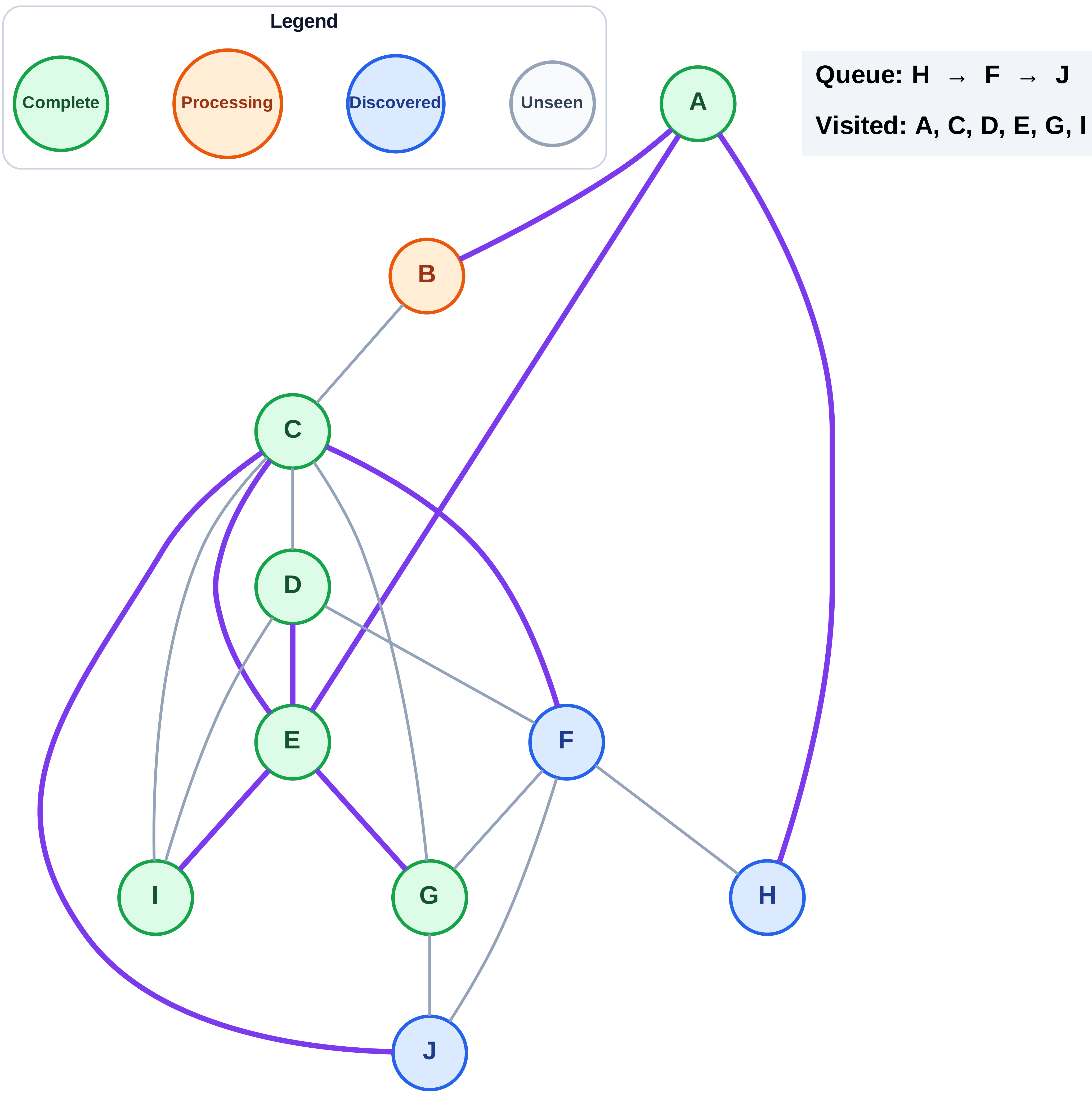
Visited: A, C, D, E, G

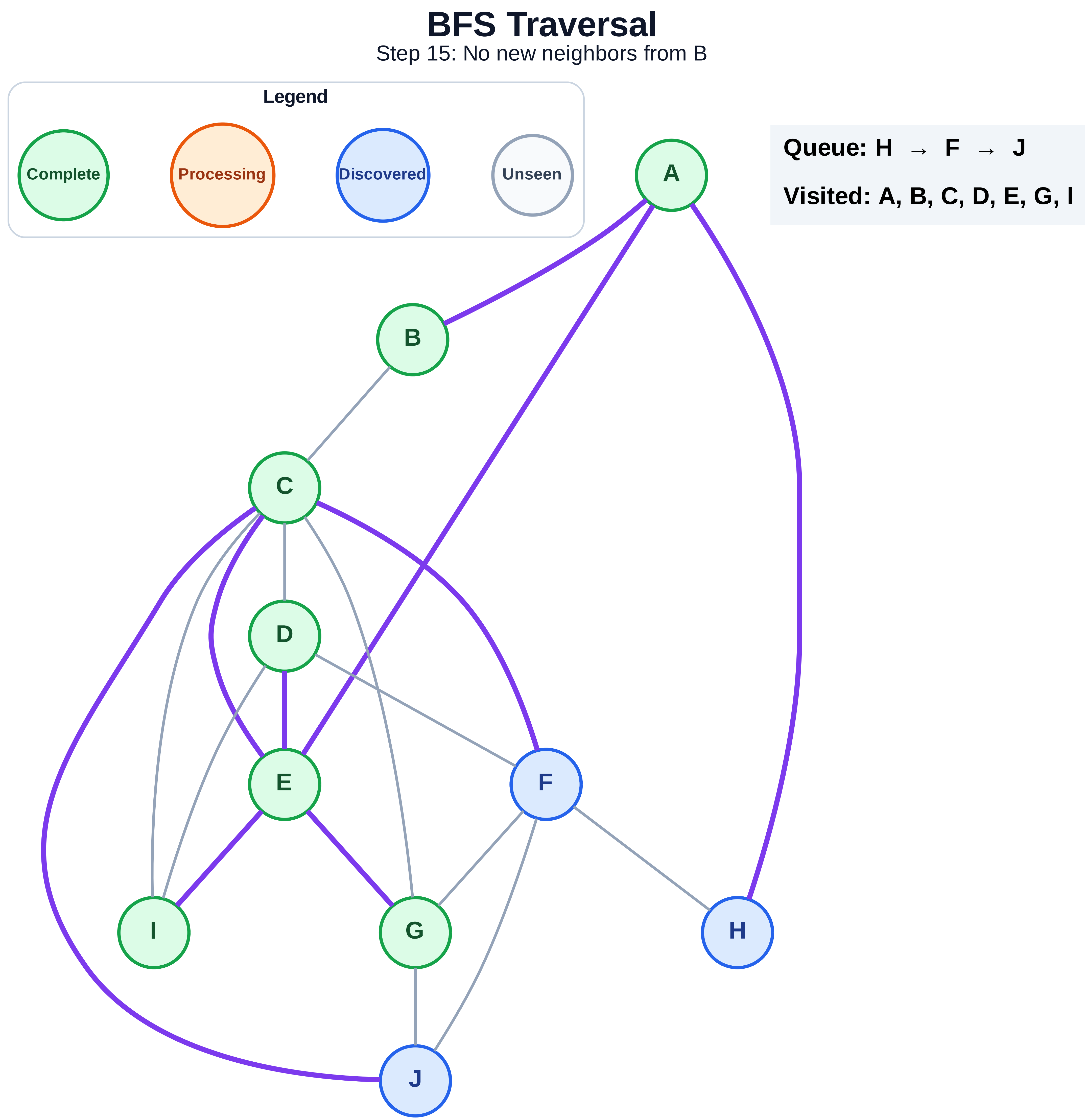


BFS Traversal
Step 13: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Queue: B → H → F → J
Visited: A, C, D, E, G, I
A
B
C
D
E
F
G
H
I
J

BFS Traversal

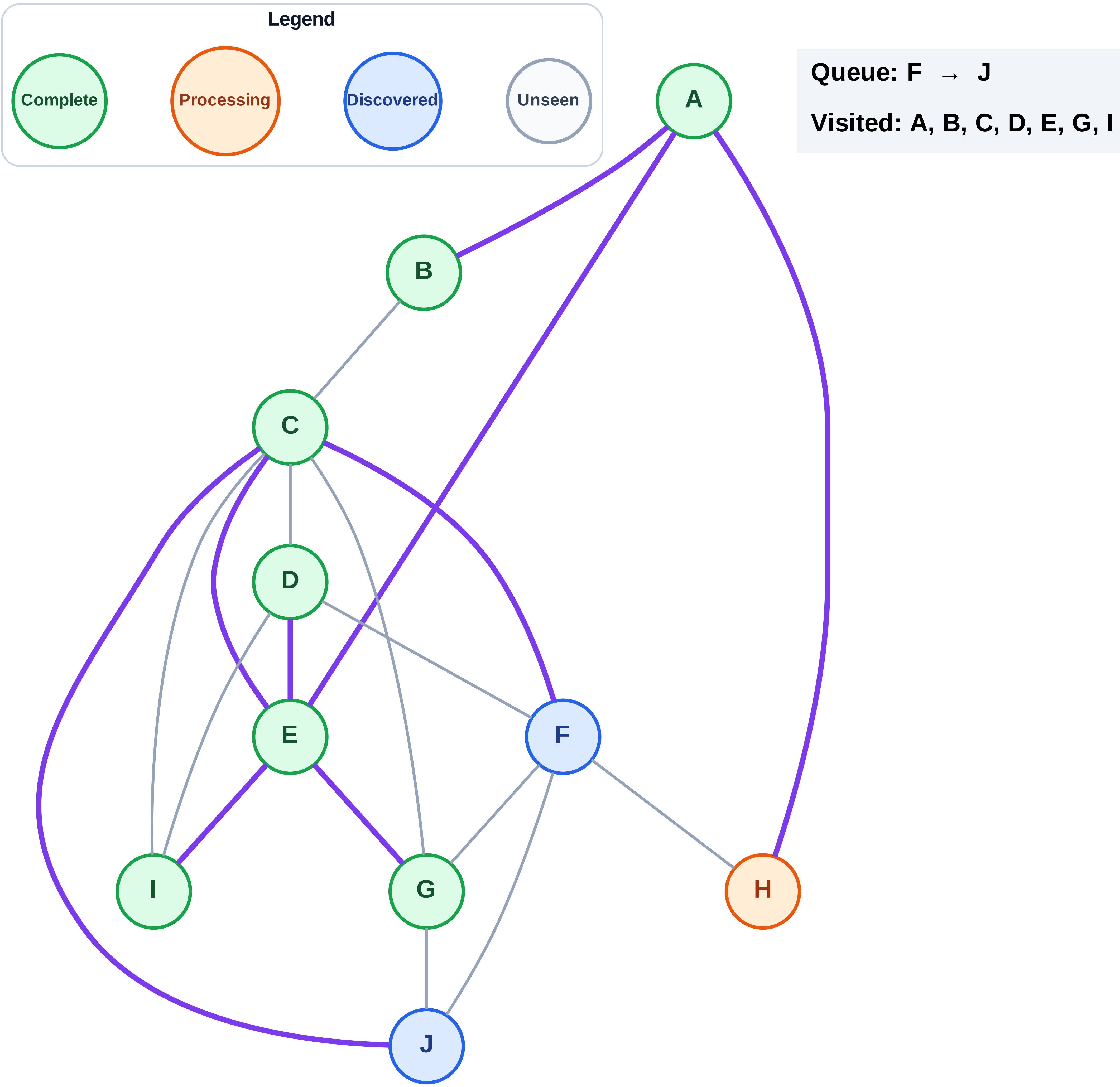
Step 14: Process node B




BFS Traversal
Step 15: No new neighbors from B
Legend
Complete
Processing
Discovered
Unseen
Queue: H → F → J
Visited: A, B, C, D, E, G, I
A
B
C
D
E
F
G
H
I
J

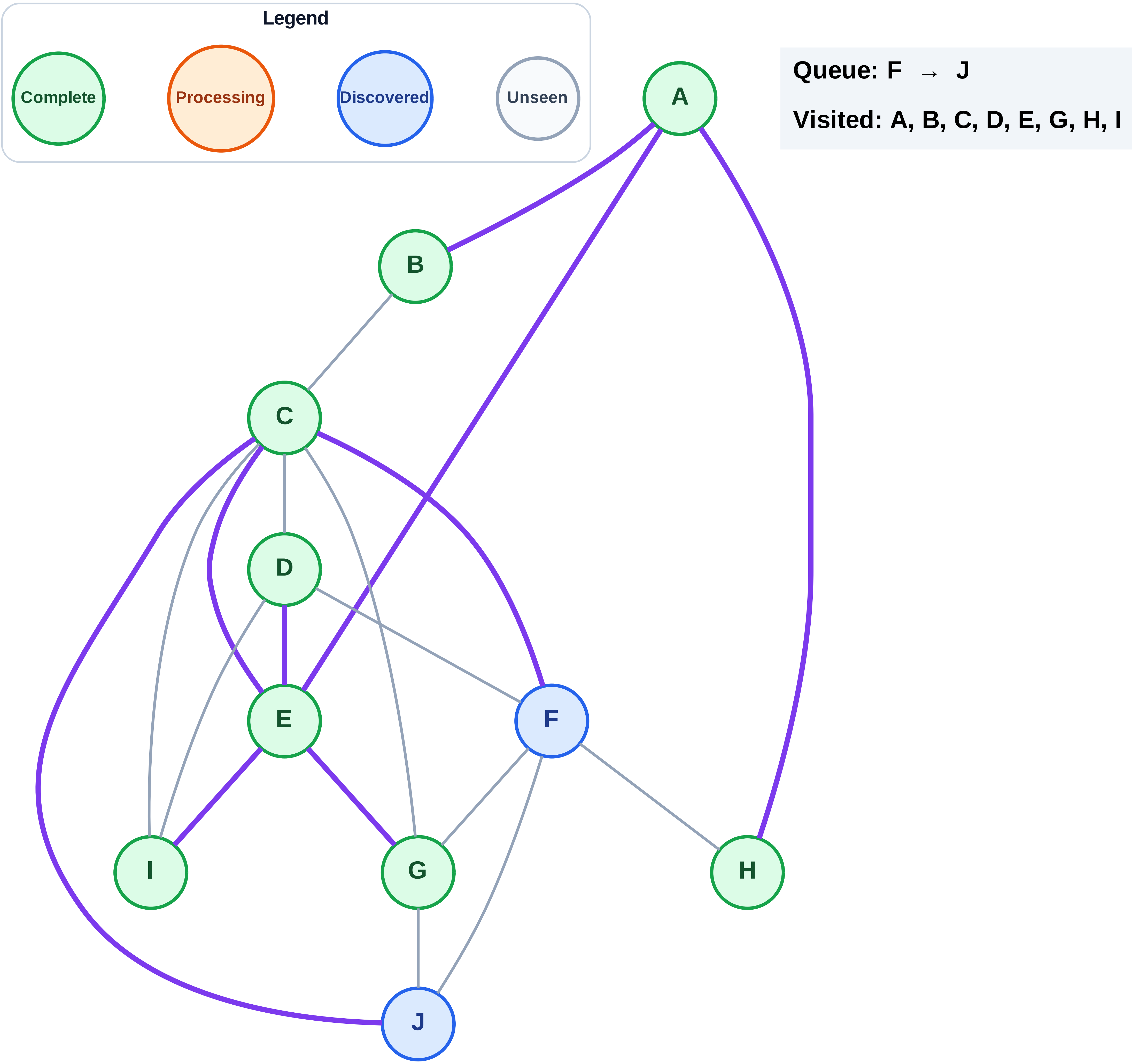
BFS Traversal

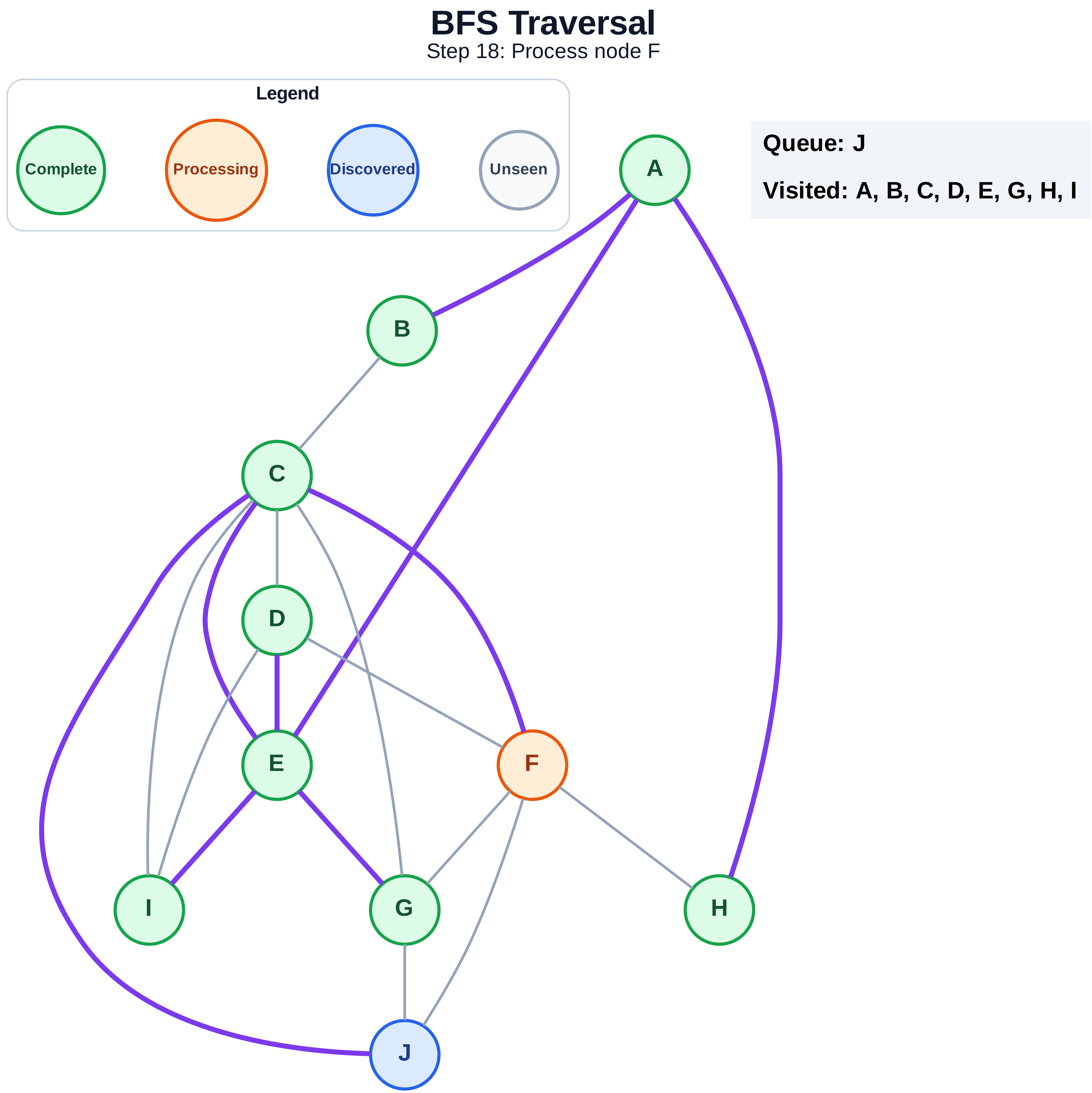
Step 16: Process node H



BFS Traversal

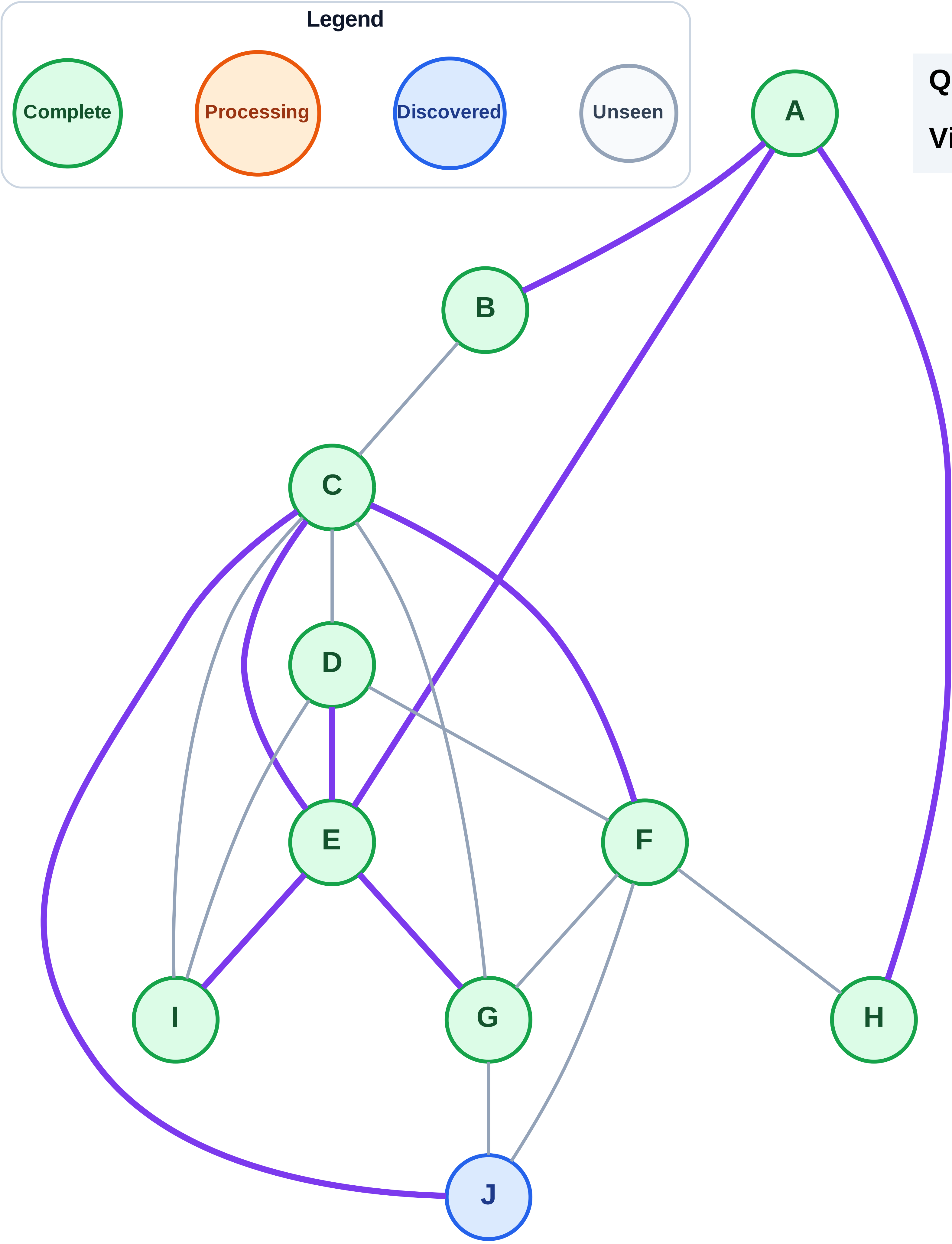
Step 17: No new neighbors from H




BFS Traversal
Step 18: Process node F
Legend
Complete
Processing
Discovered
Unseen
Queue: J
Visited: A, B, C, D, E, G, H, I
A
B
C
D
E
F
G
H
I
J

BFS Traversal

Step 19: No new neighbors from F

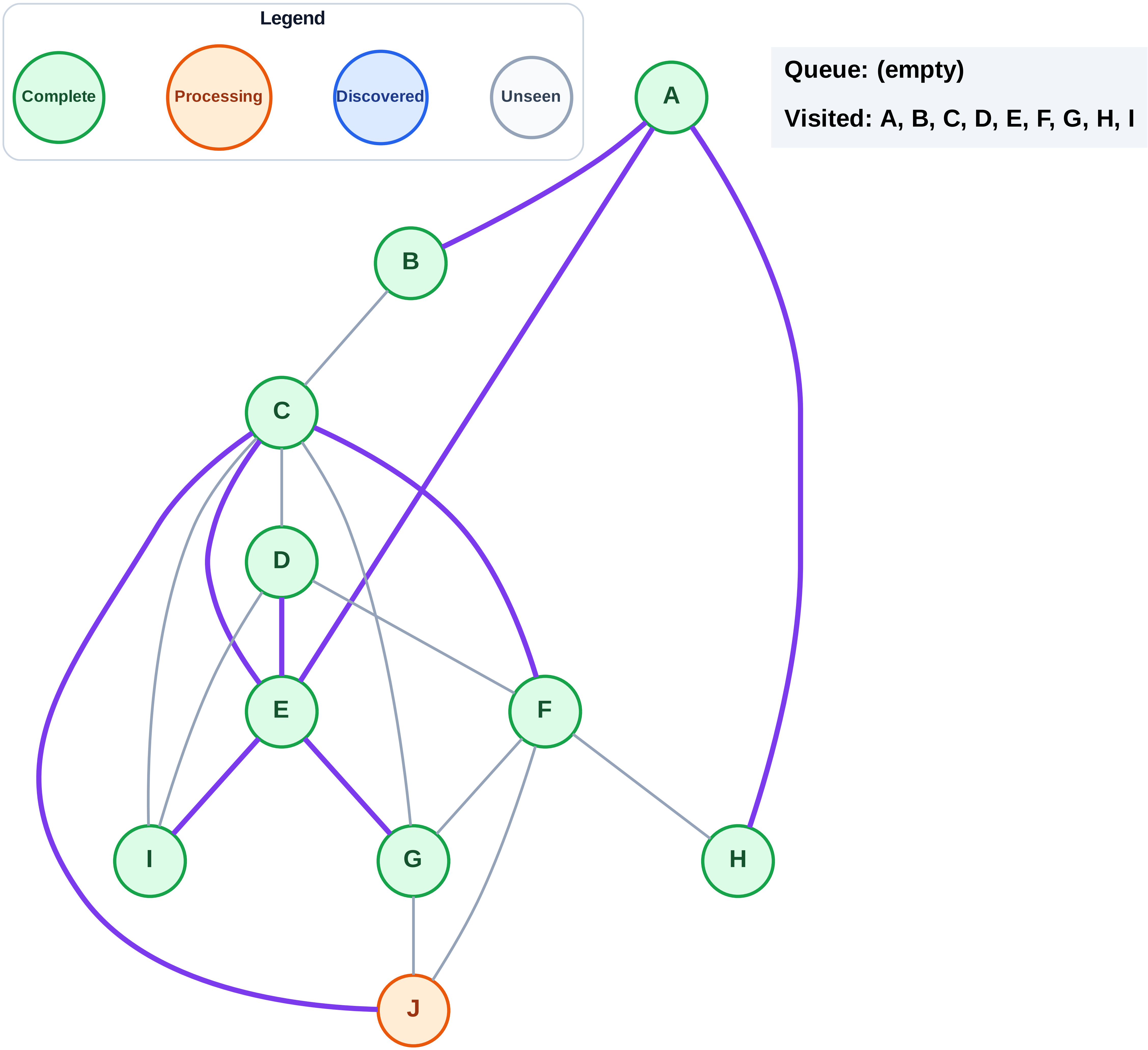


Queue: J

Visited: A, B, C, D, E, F, G, H, I

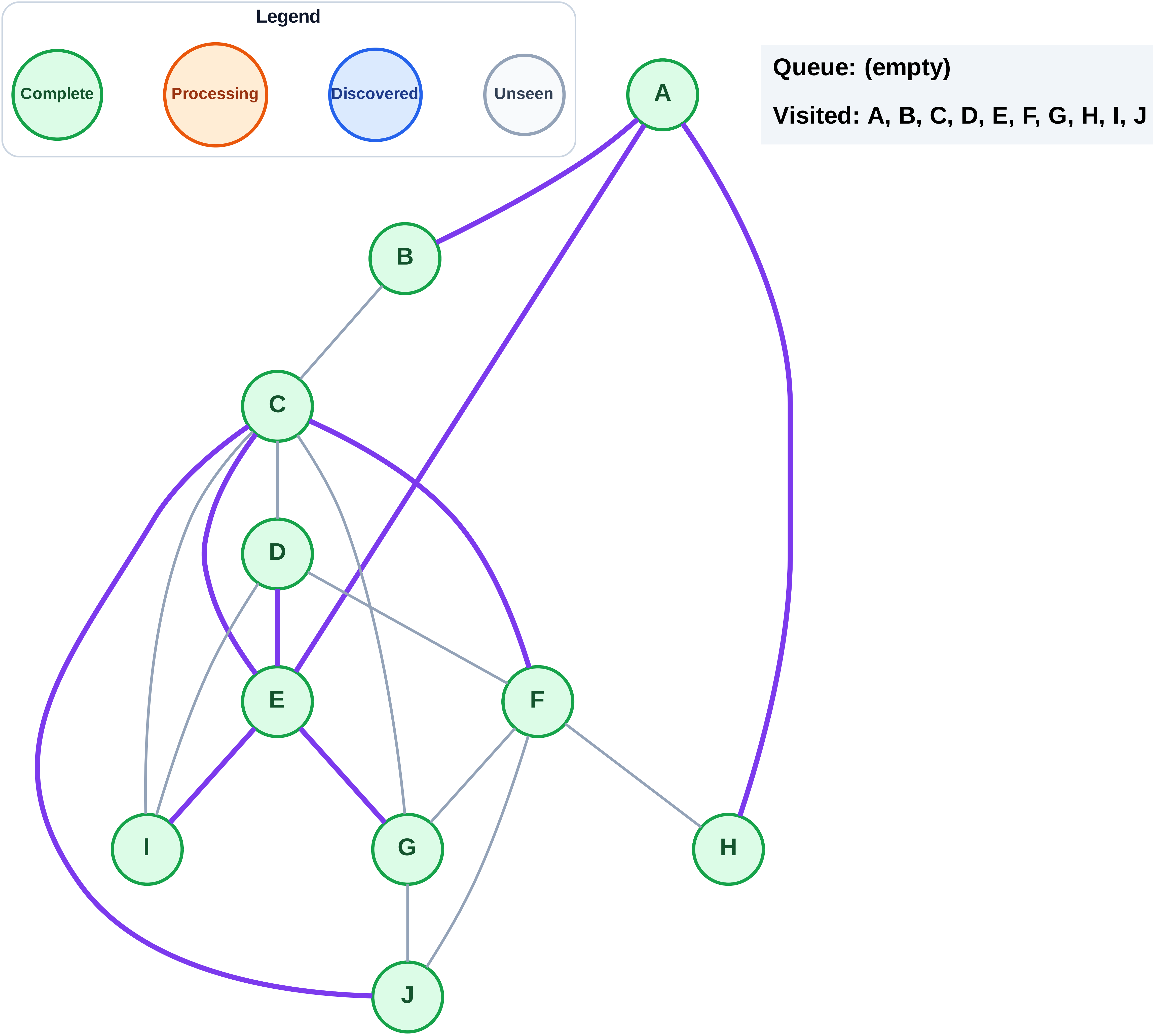
BFS Traversal

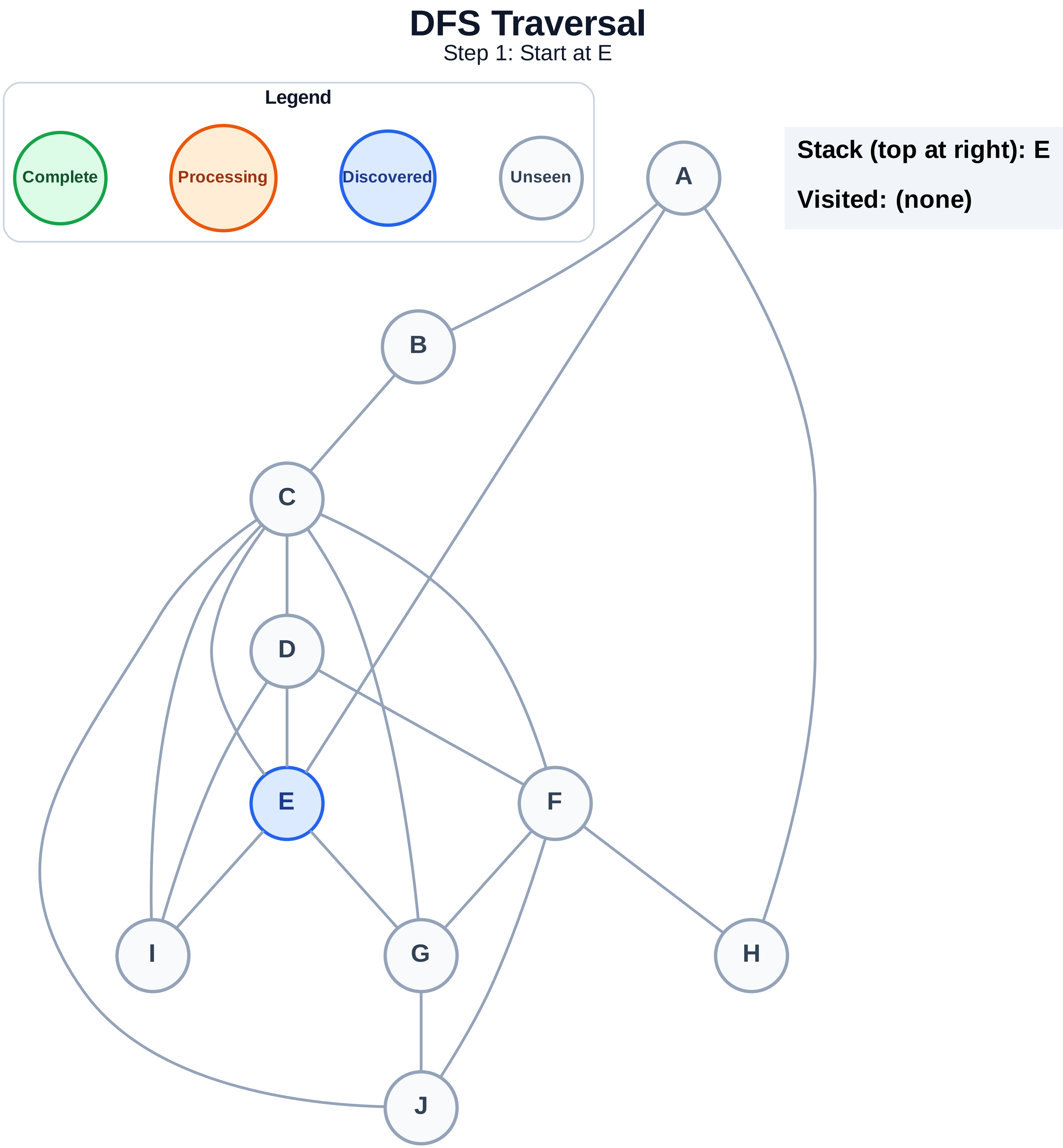
Step 20: Process node J



BFS Traversal

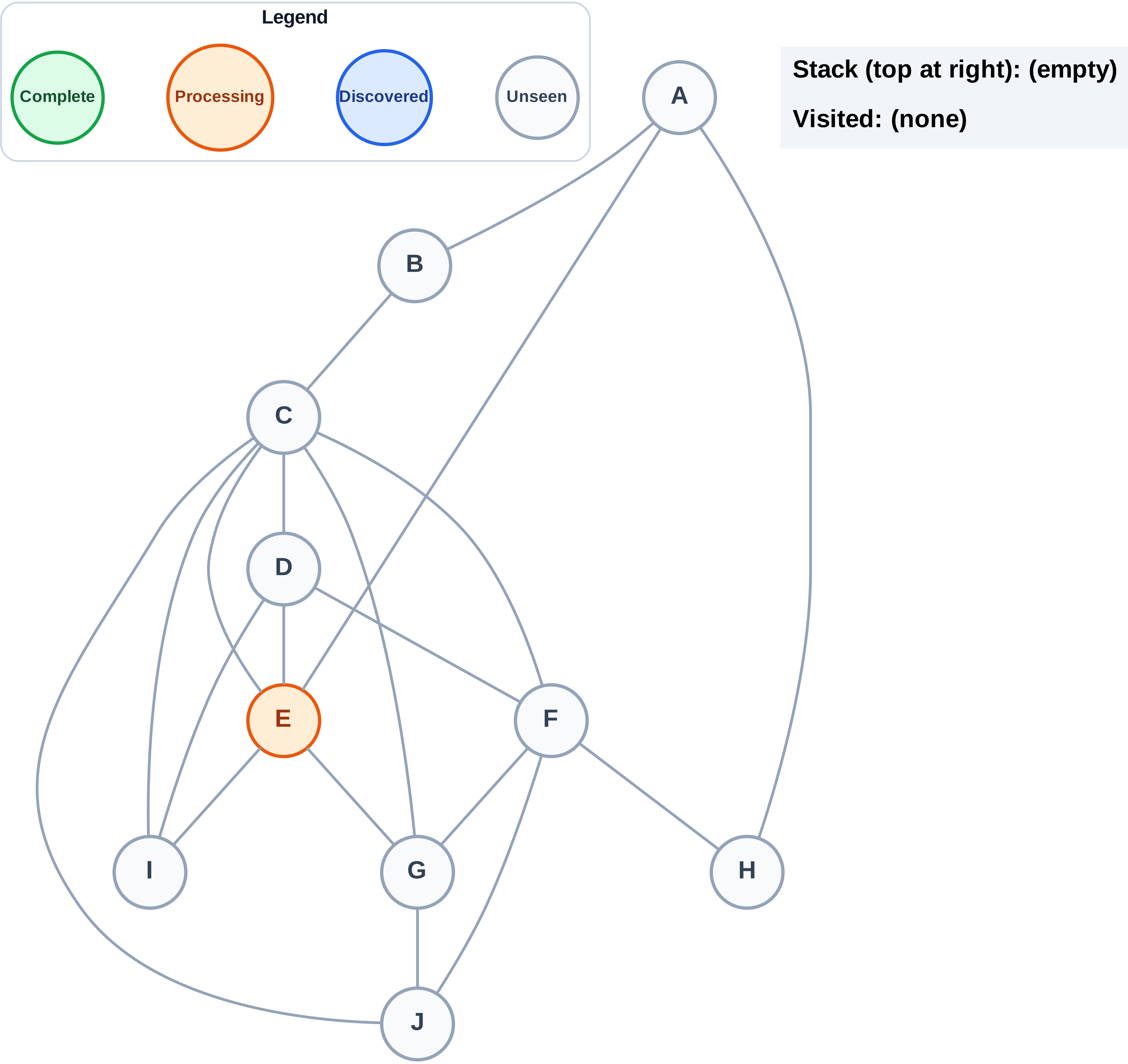
Step 21: No new neighbors from J




DFS Traversal
Step 1: Start at E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): E
Visited: (none)
A
B
C
D
E
F
G
H
I
J

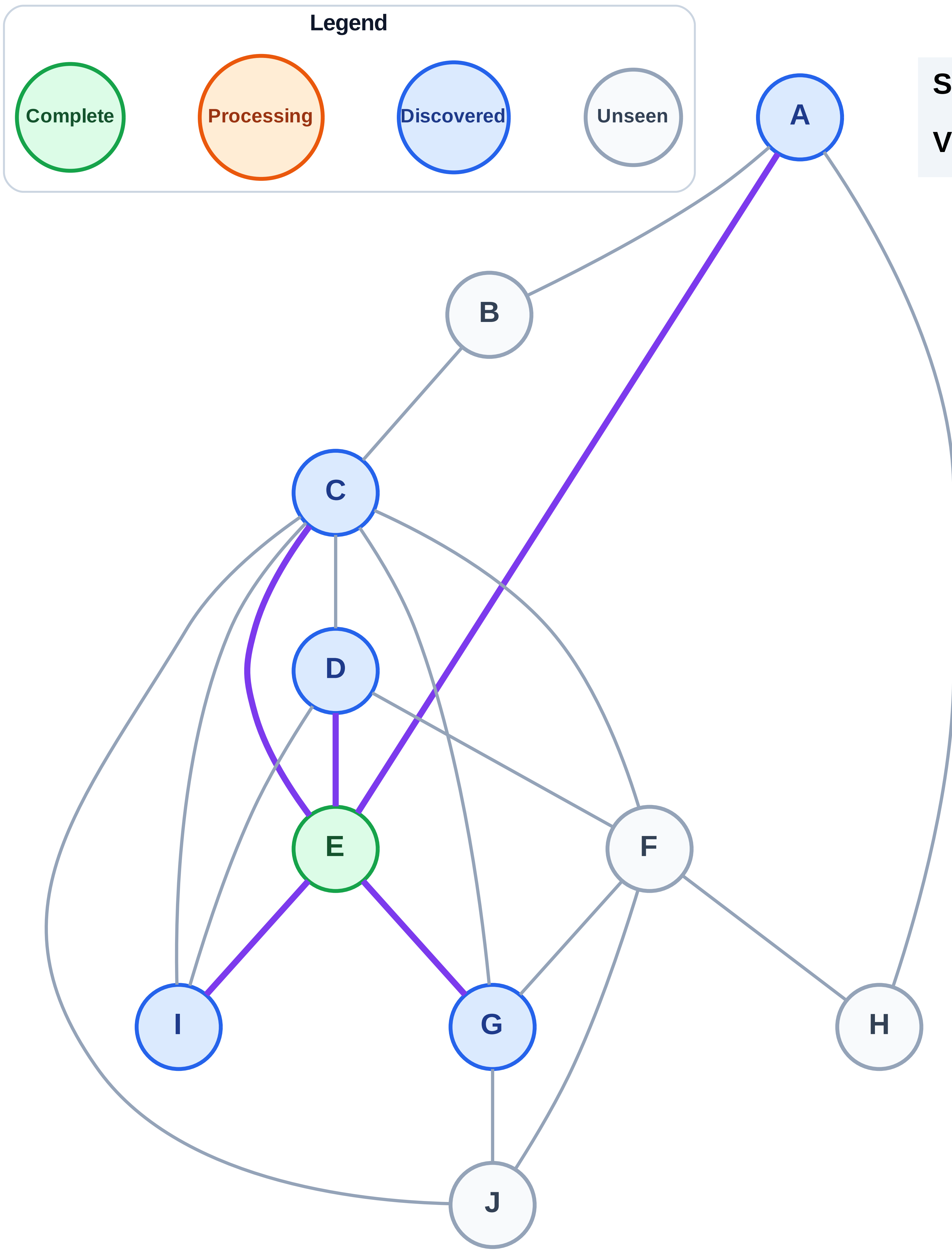
DFS Traversal

Step 2: Process node E



DFS Traversal

Step 3: Discover I, G, D, C, A from E

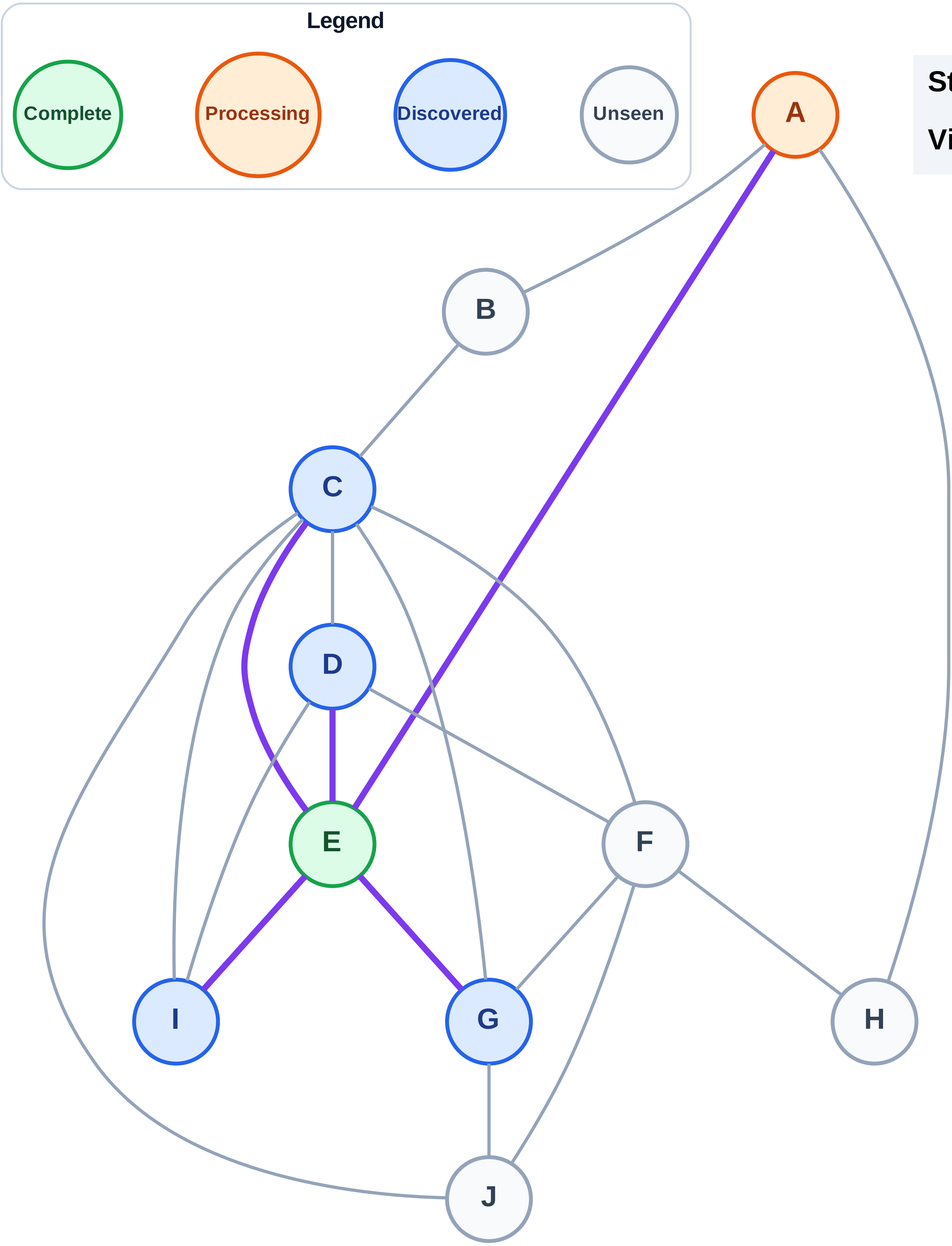


Stack (top at right): I | G | D | C | A

Visited: E

DFS Traversal

Step 4: Process node A

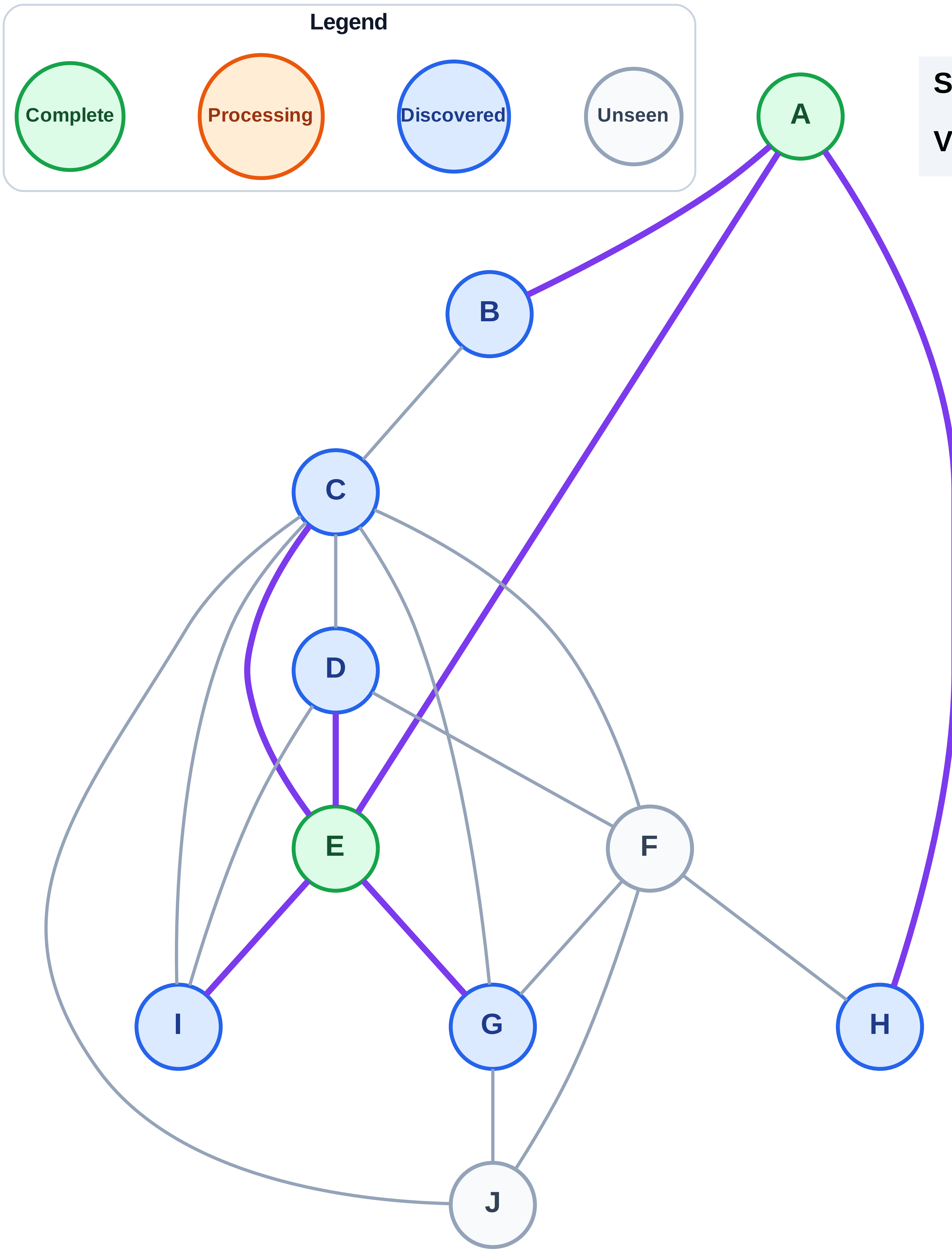


Stack (top at right): I | G | D | C

Visited: E

DFS Traversal

Step 5: Discover H, B from A

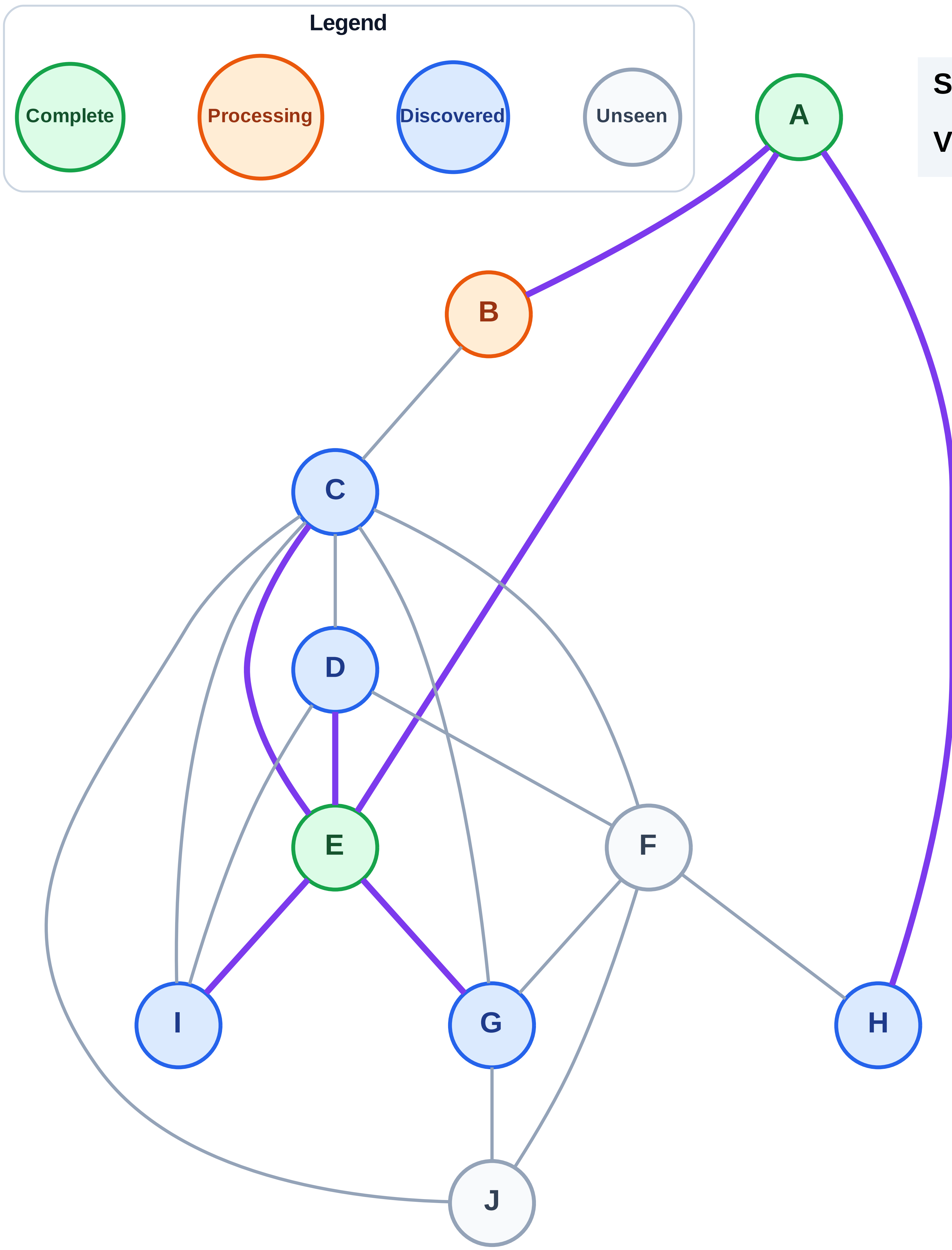


Stack (top at right): I | G | D | C | H | B

Visited: A, E

DFS Traversal

Step 6: Process node B

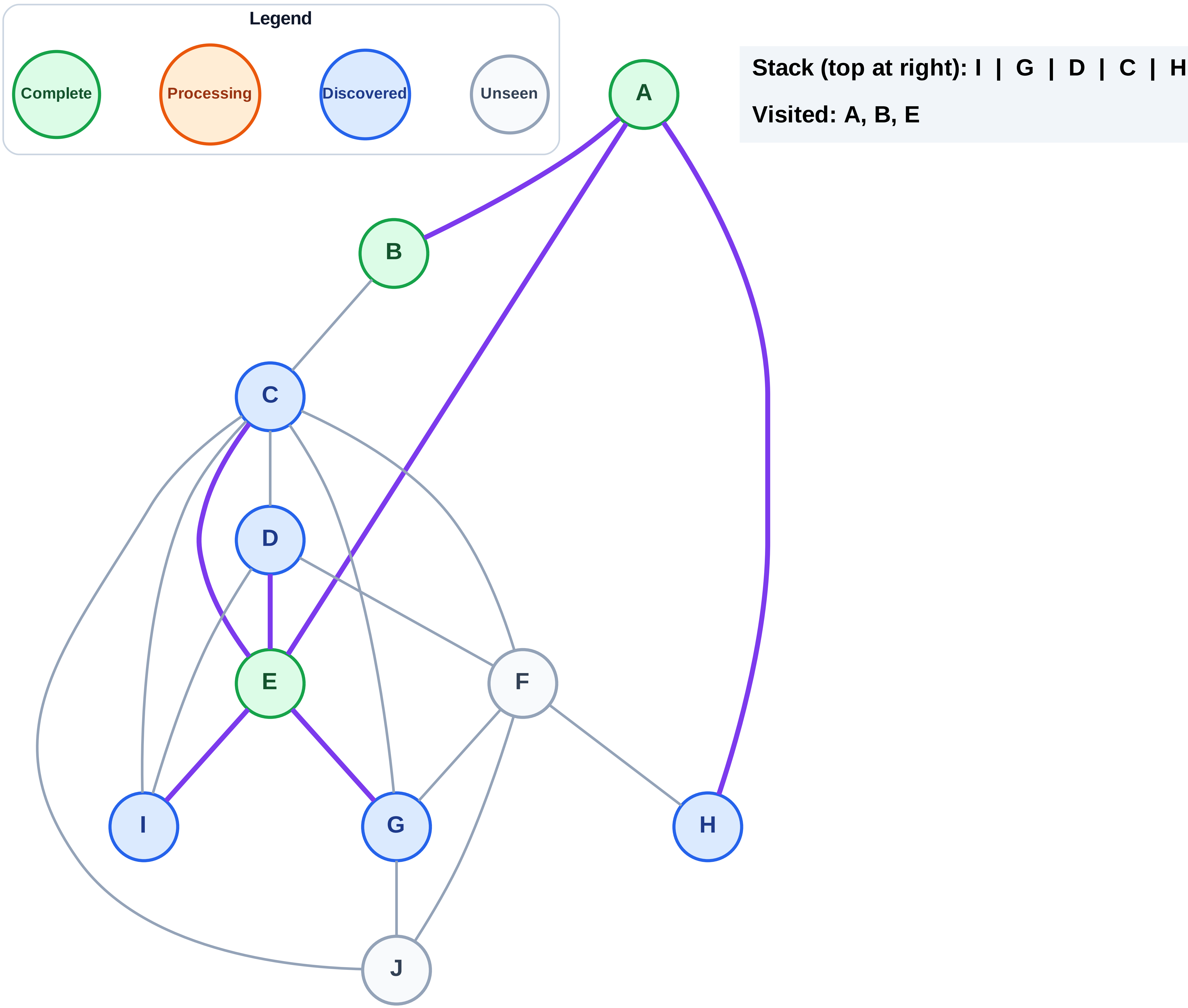


Stack (top at right): I | G | D | C | H

Visited: A, E

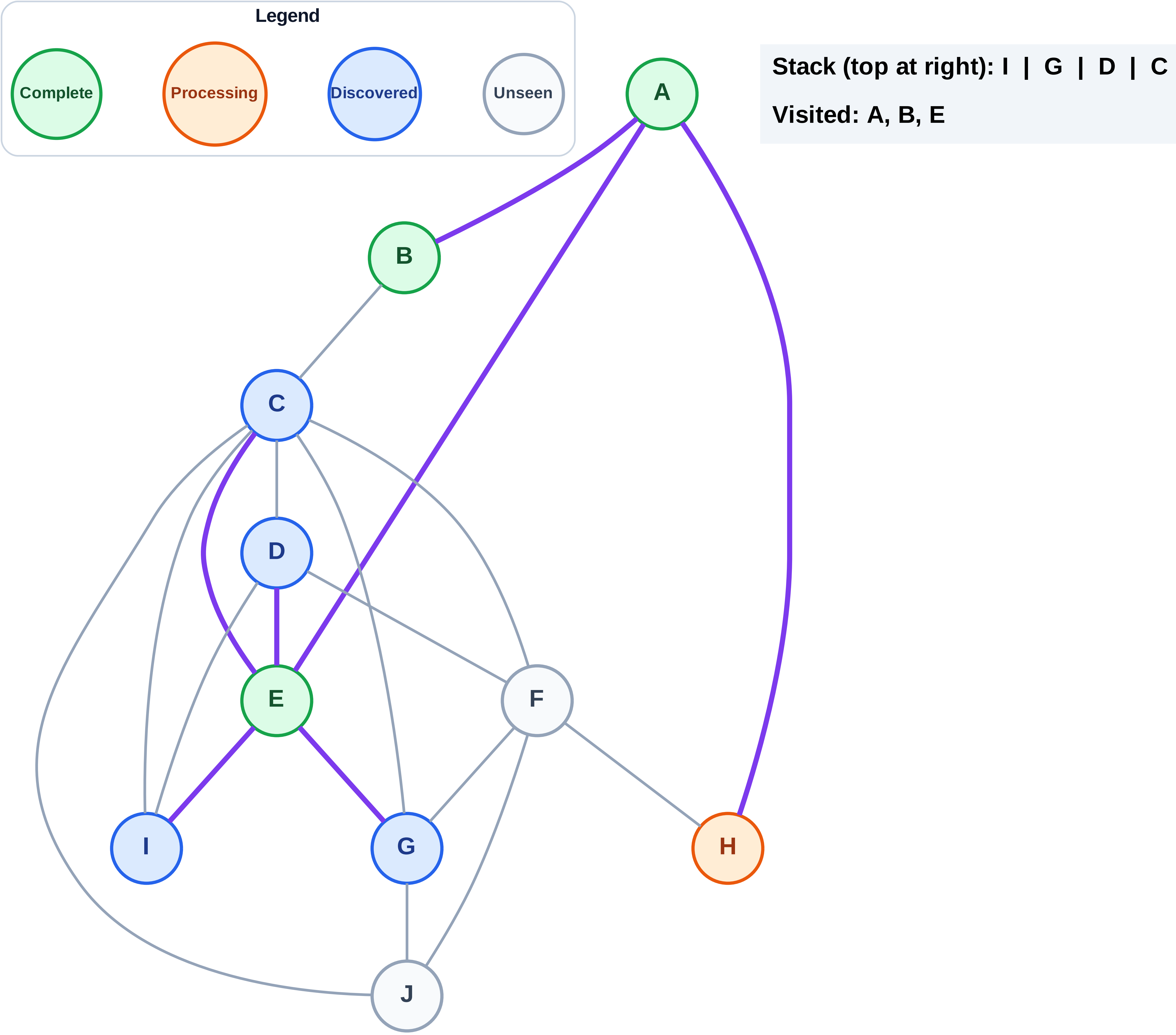
DFS Traversal

Step 7: No new neighbors from B



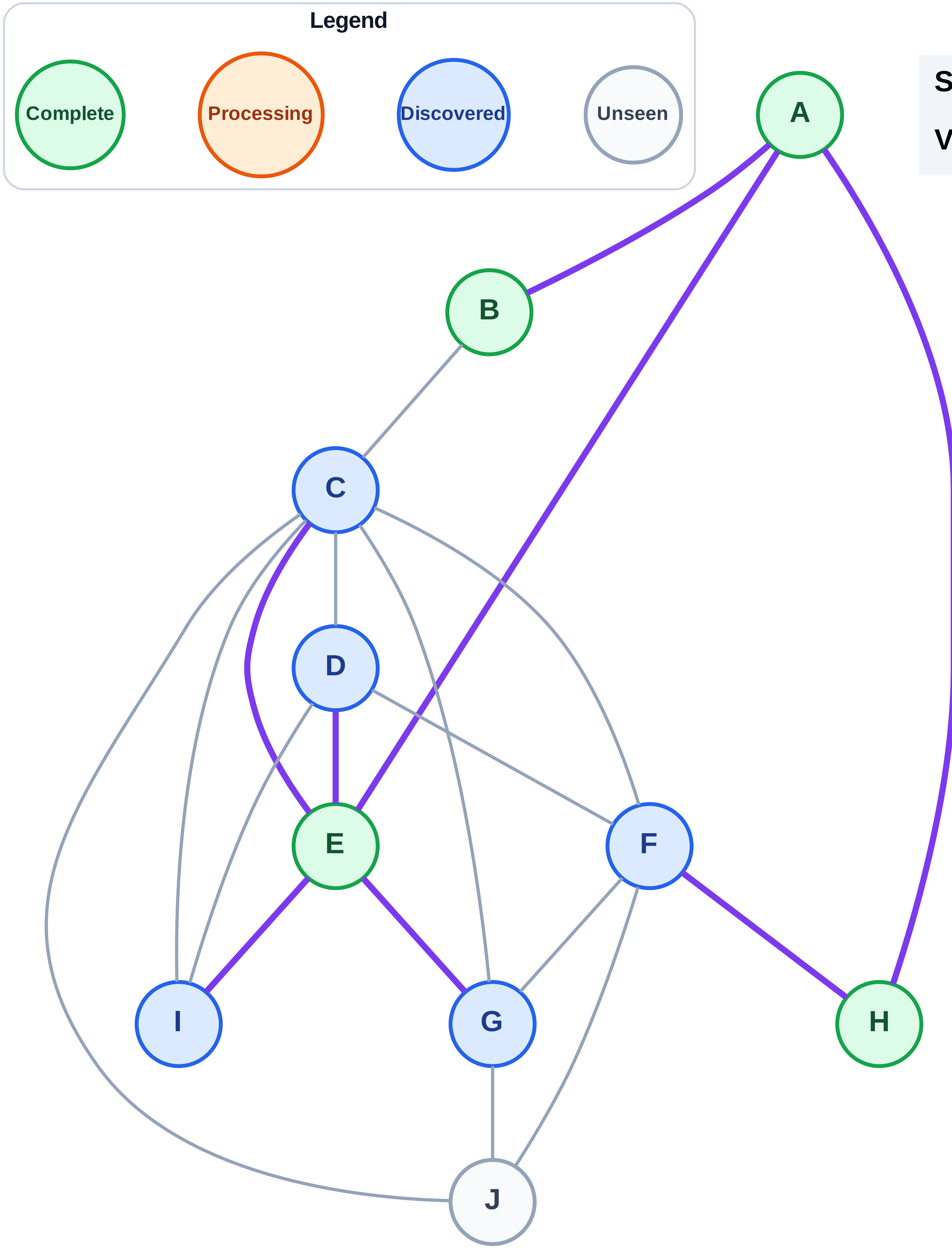
DFS Traversal

Step 8: Process node H



DFS Traversal

Step 9: Discover F from H

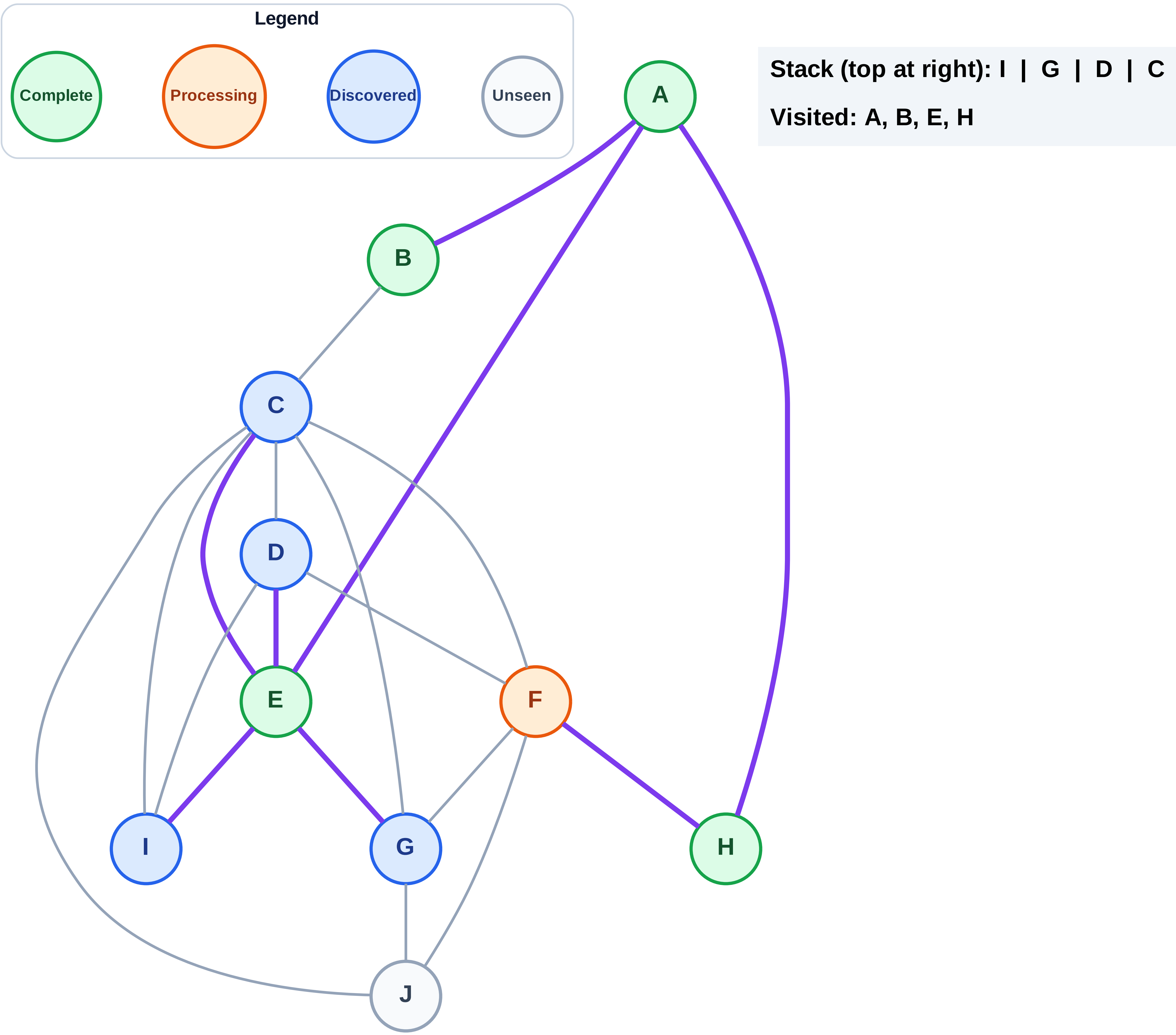


Stack (top at right): I | G | D | C | F

Visited: A, B, E, H

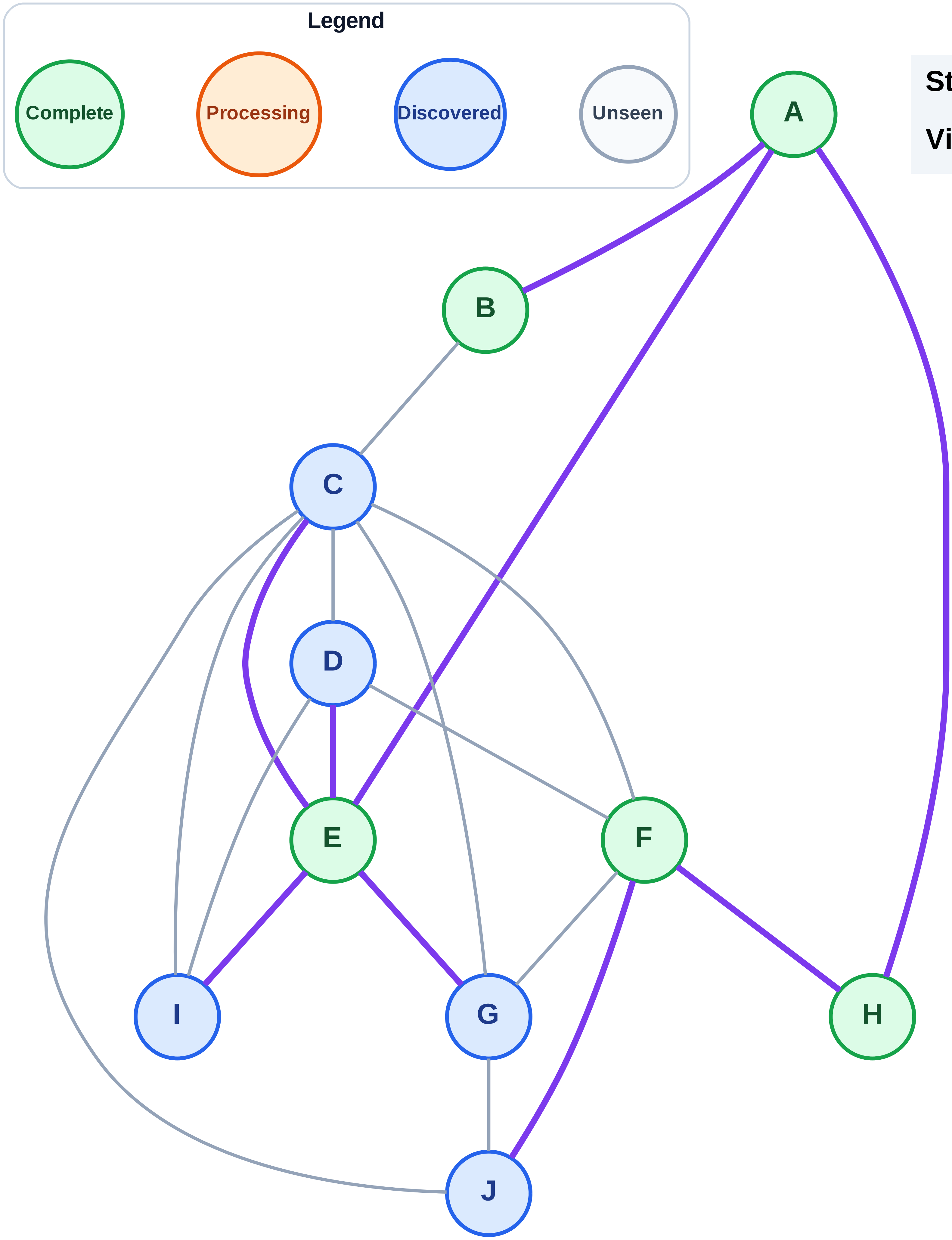
DFS Traversal

Step 10: Process node F



DFS Traversal

Step 11: Discover J from F

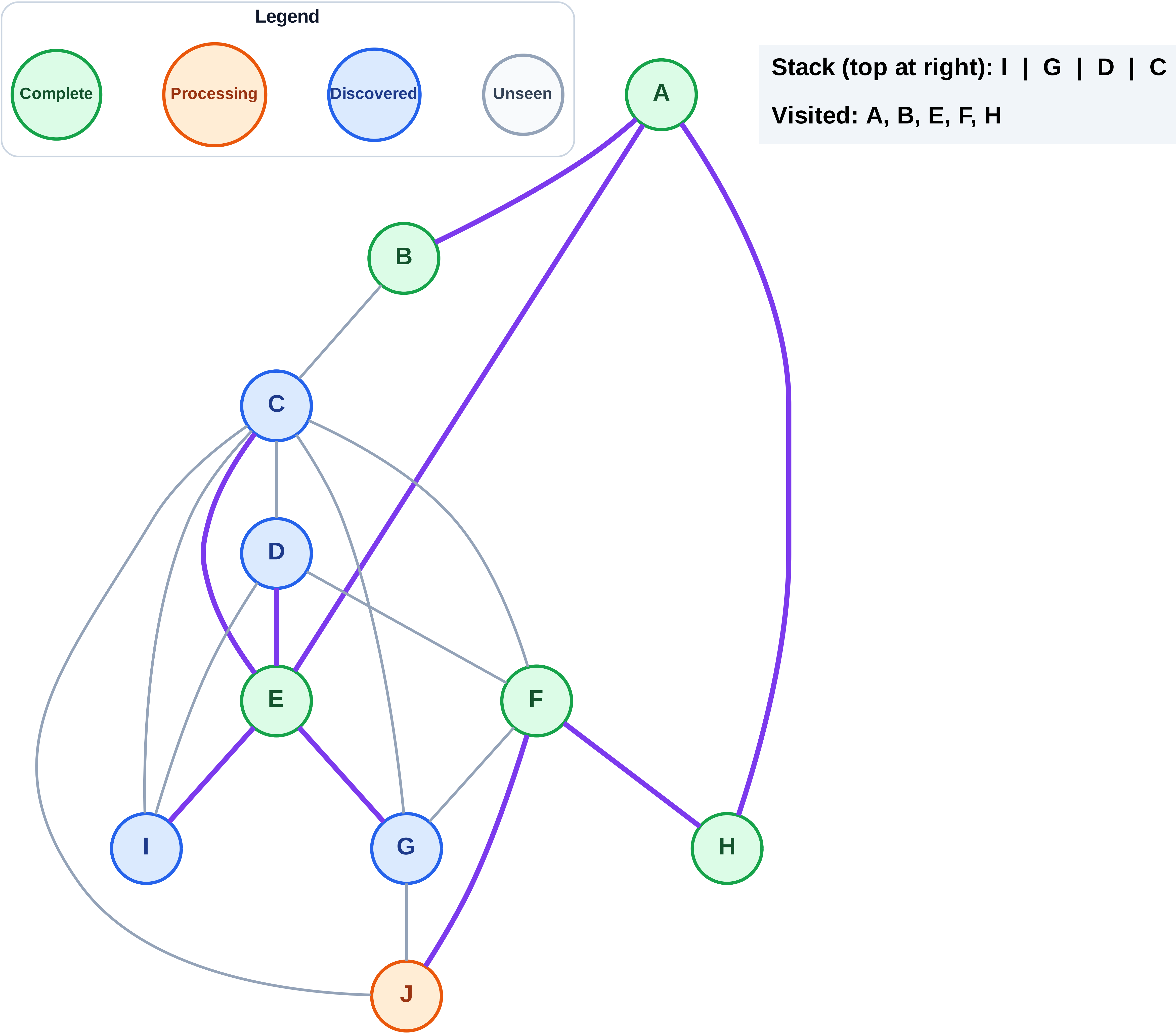


Stack (top at right): I | G | D | C | J

Visited: A, B, E, F, H

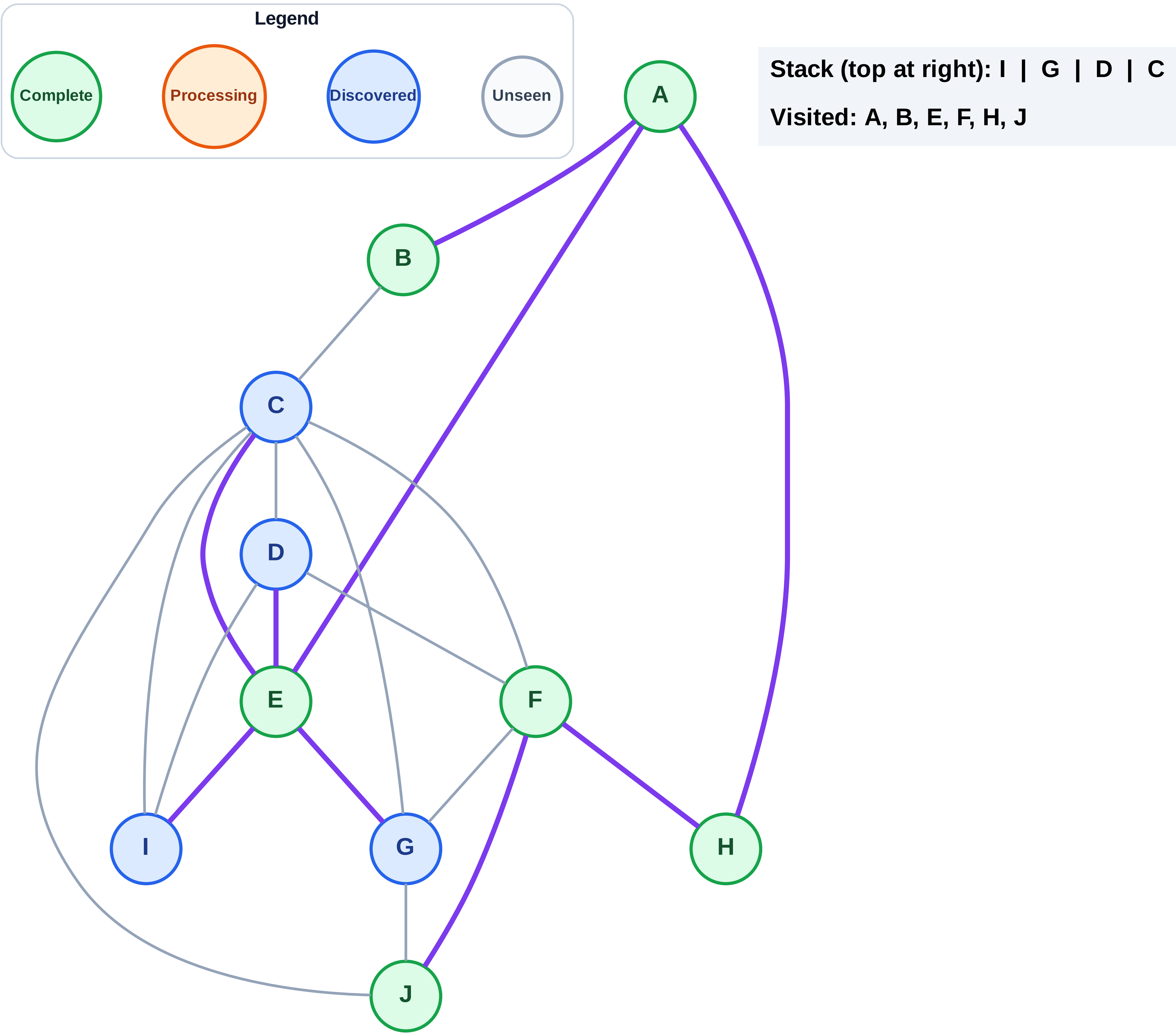
DFS Traversal

Step 12: Process node J



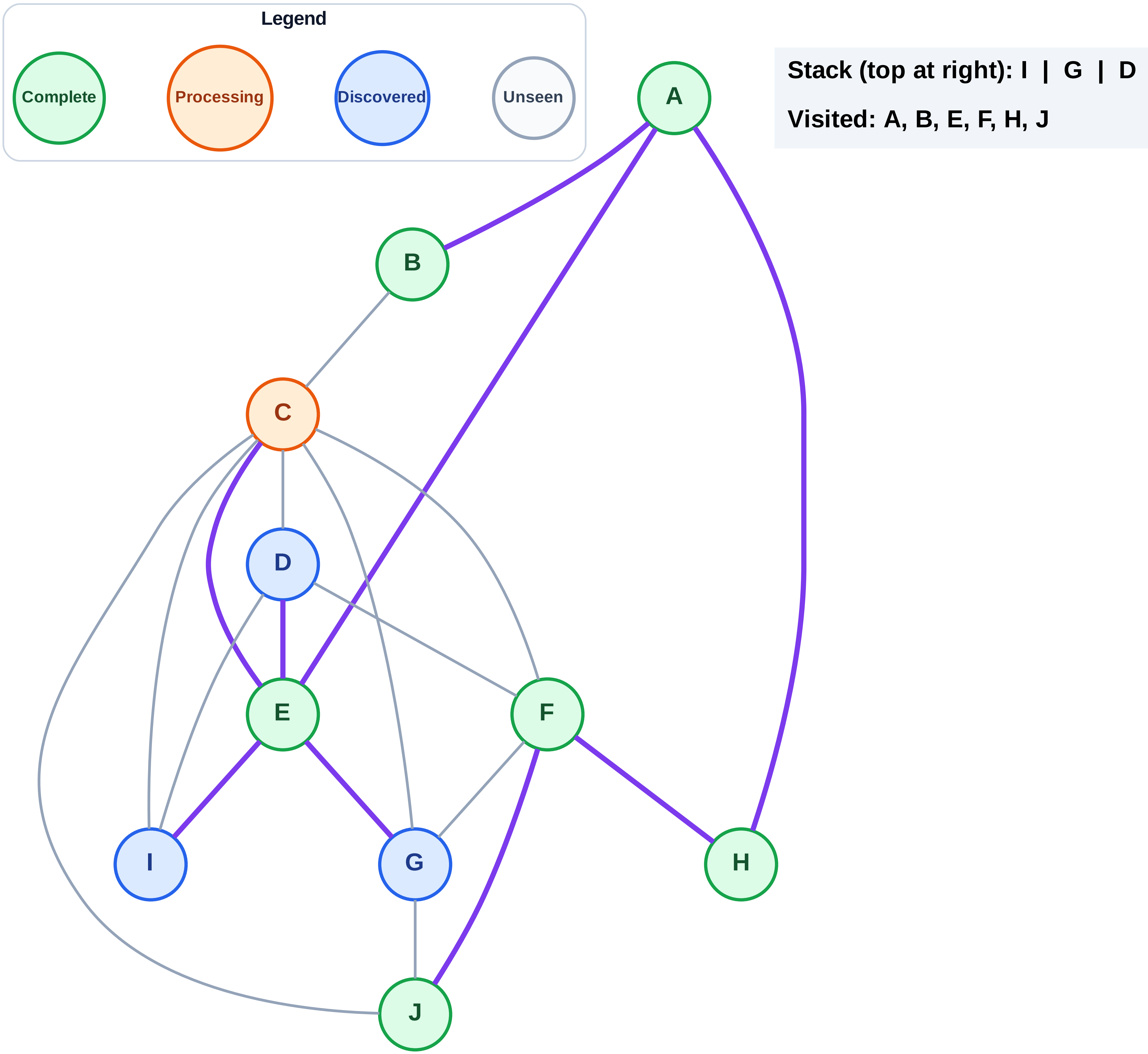
DFS Traversal

Step 13: No new neighbors from J



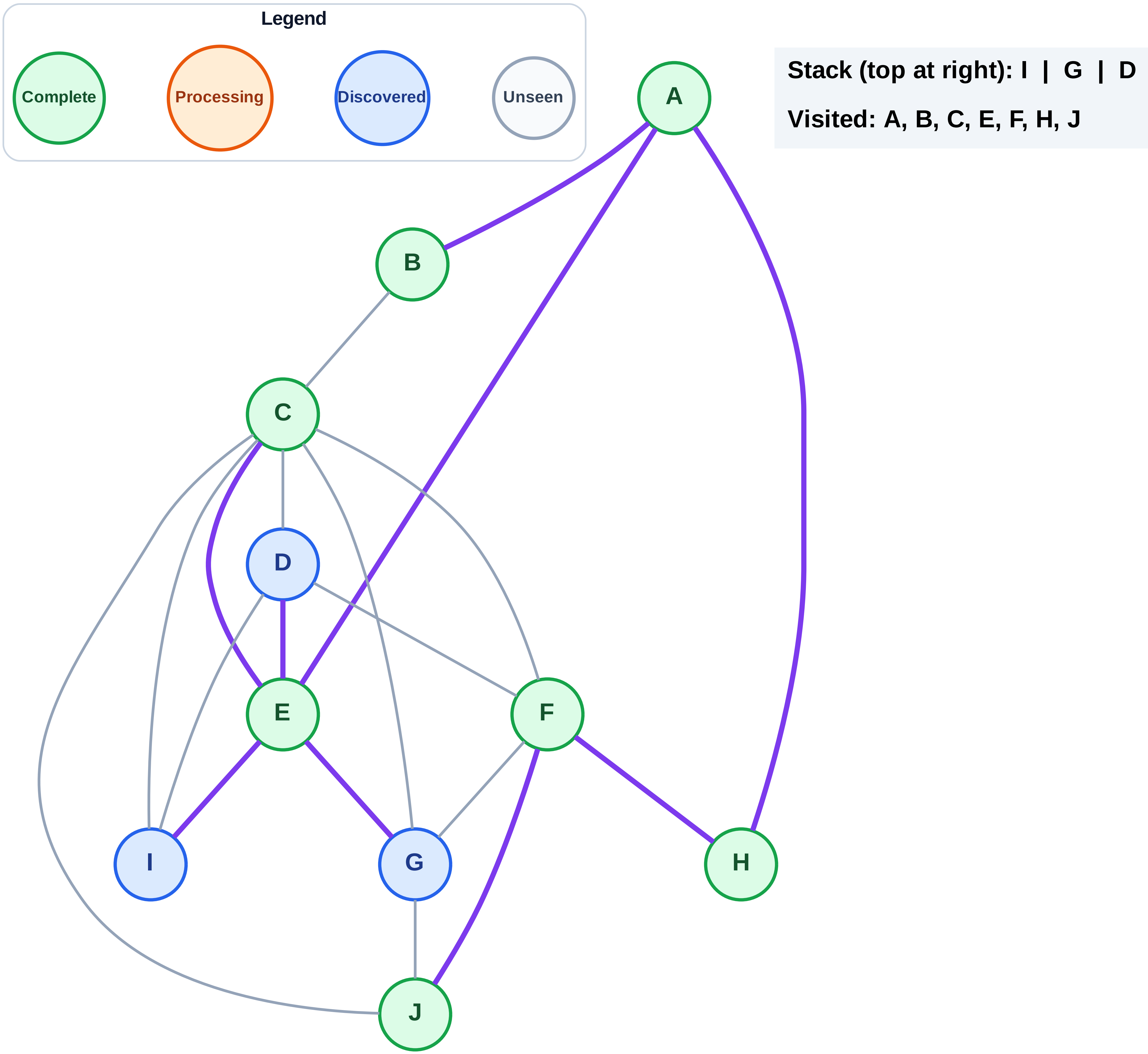
DFS Traversal

Step 14: Process node C



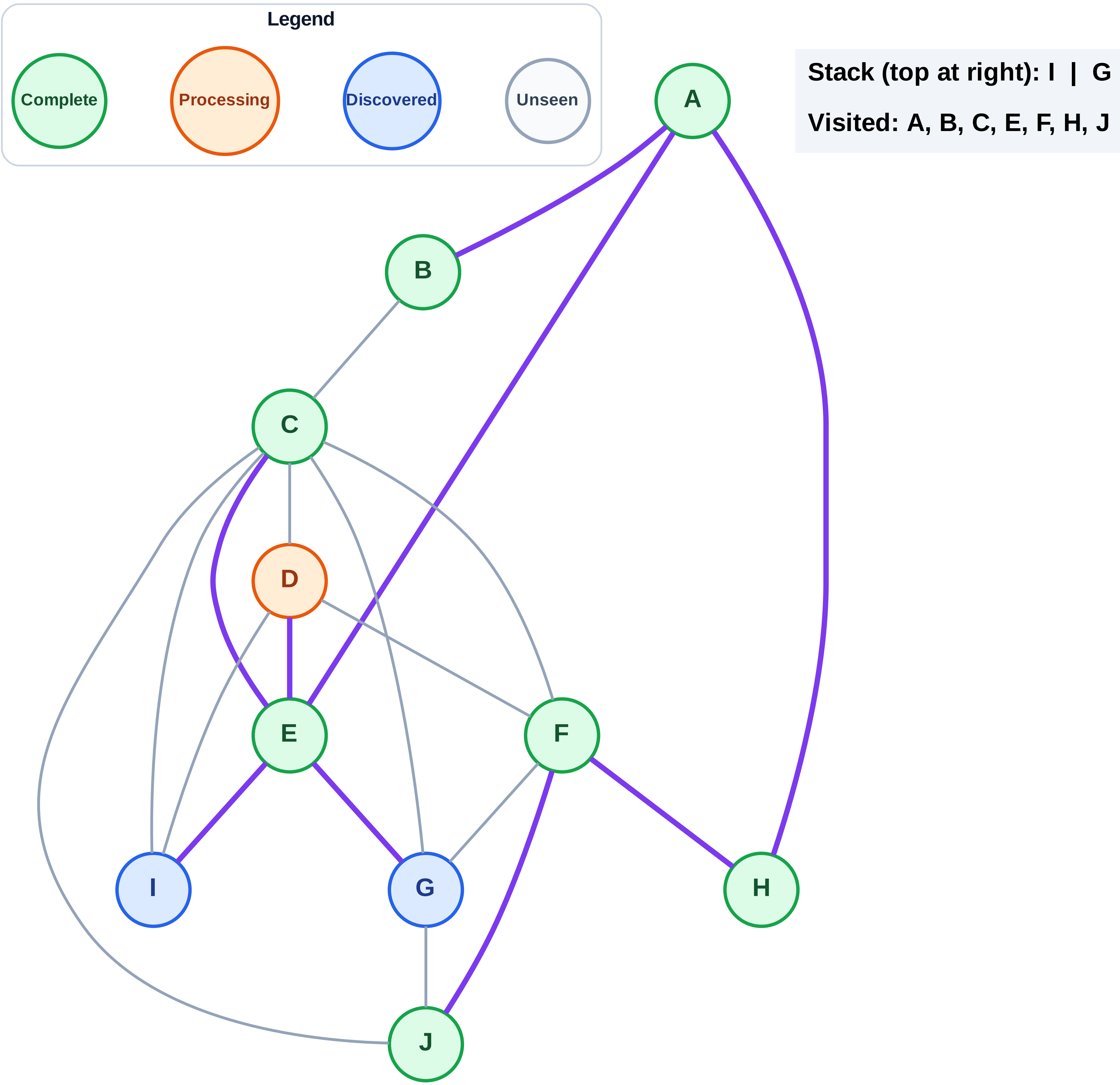
DFS Traversal

Step 15: No new neighbors from C



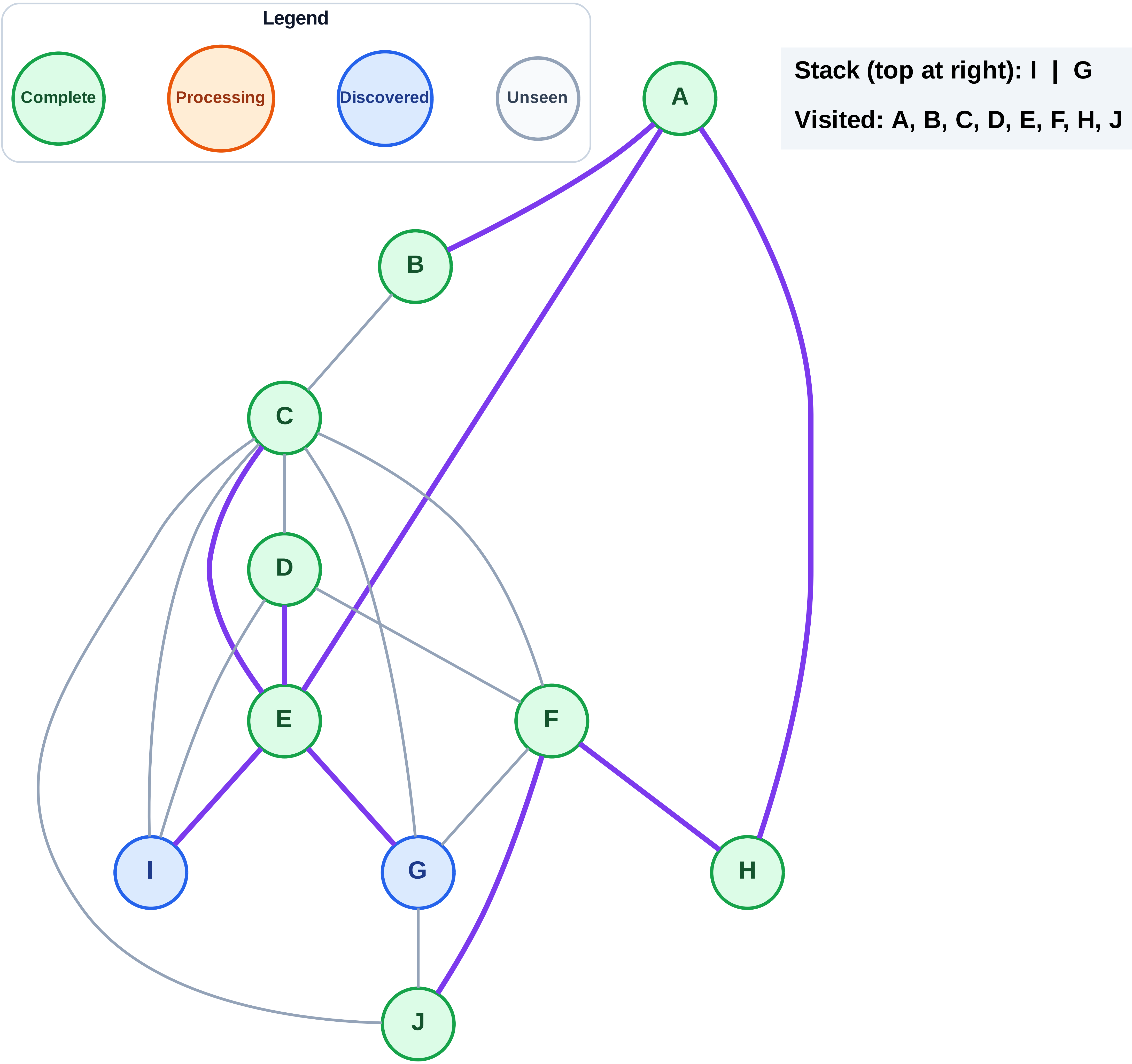
DFS Traversal

Step 16: Process node D



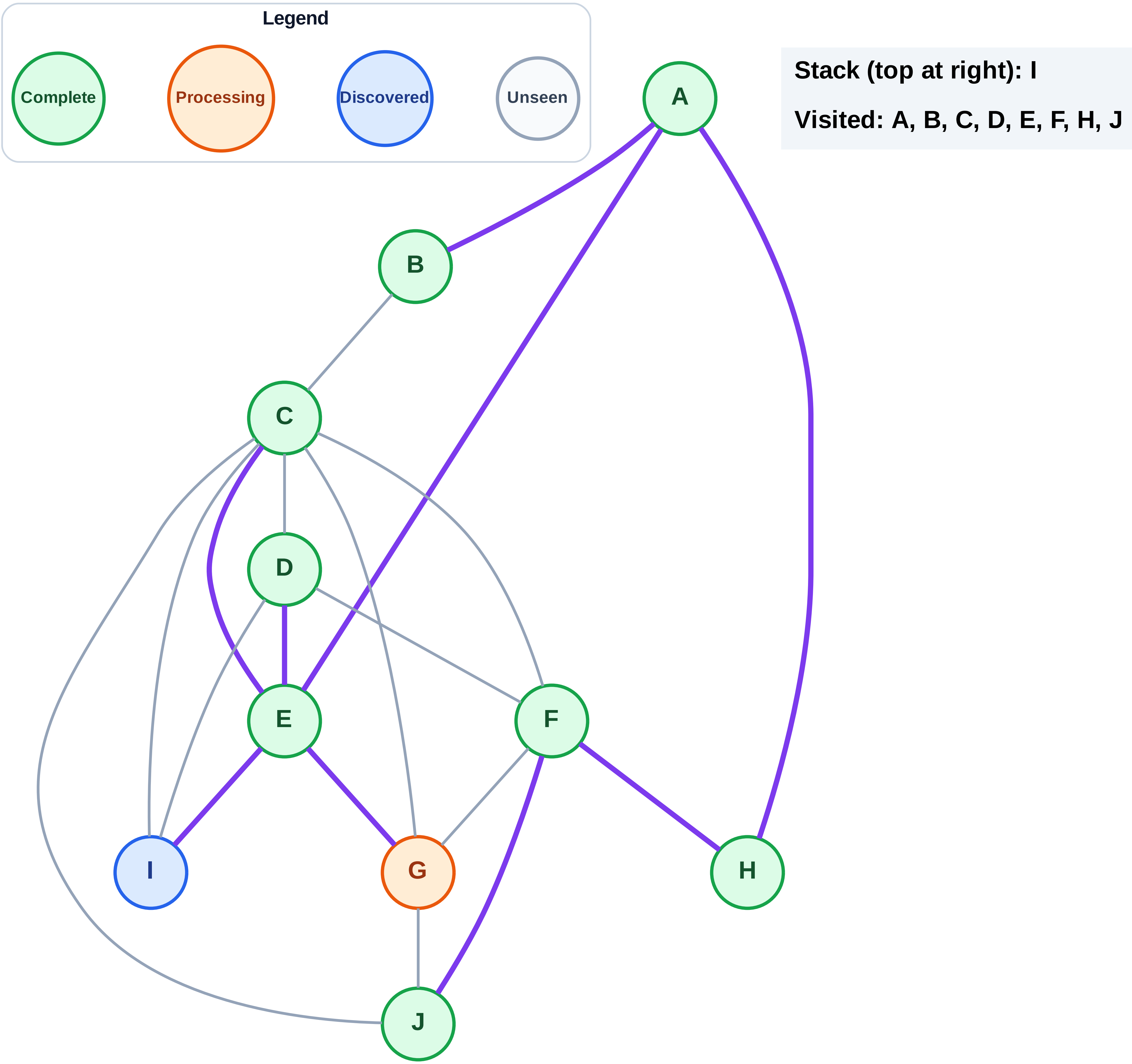
DFS Traversal

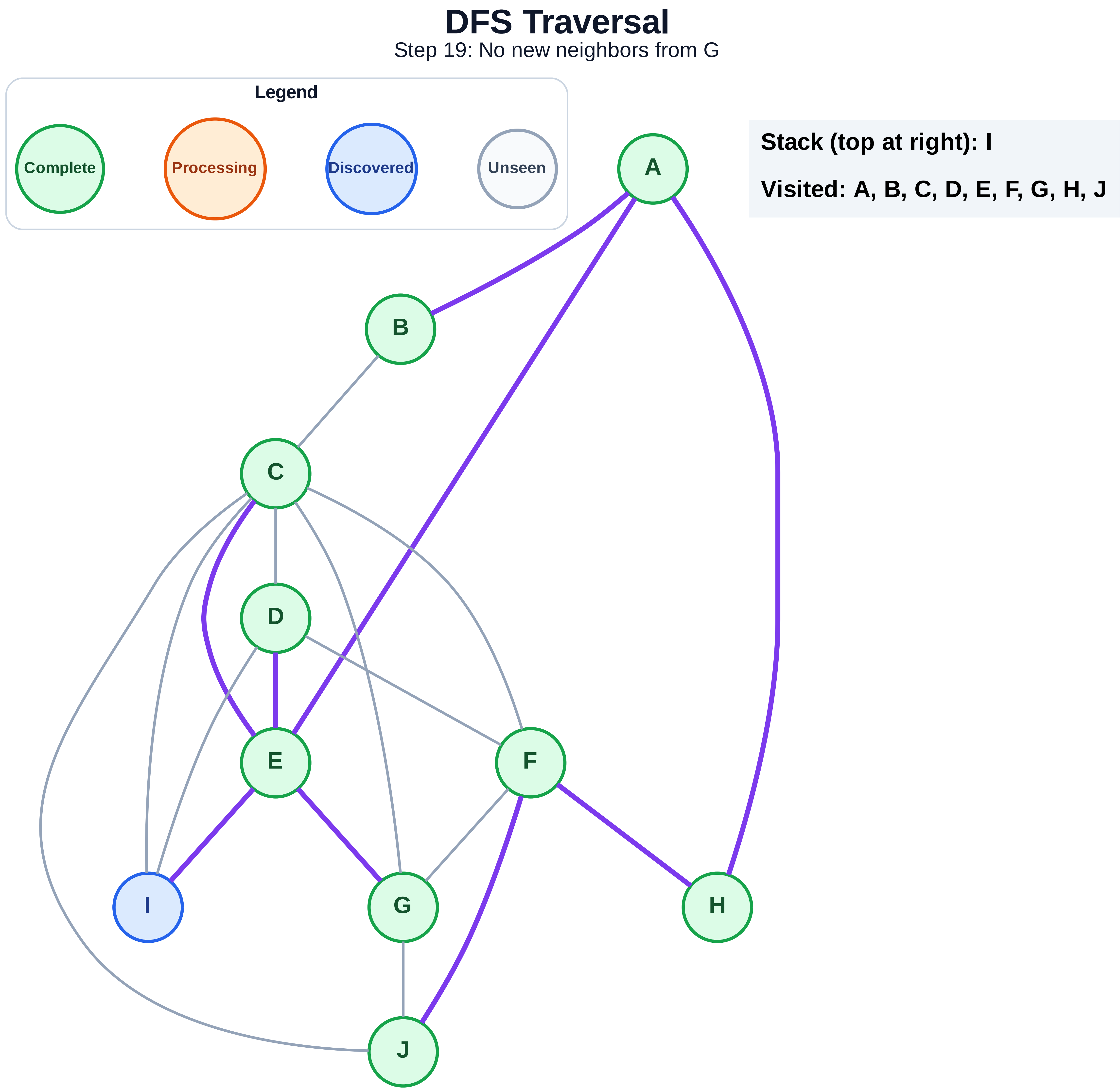
Step 17: No new neighbors from D



DFS Traversal

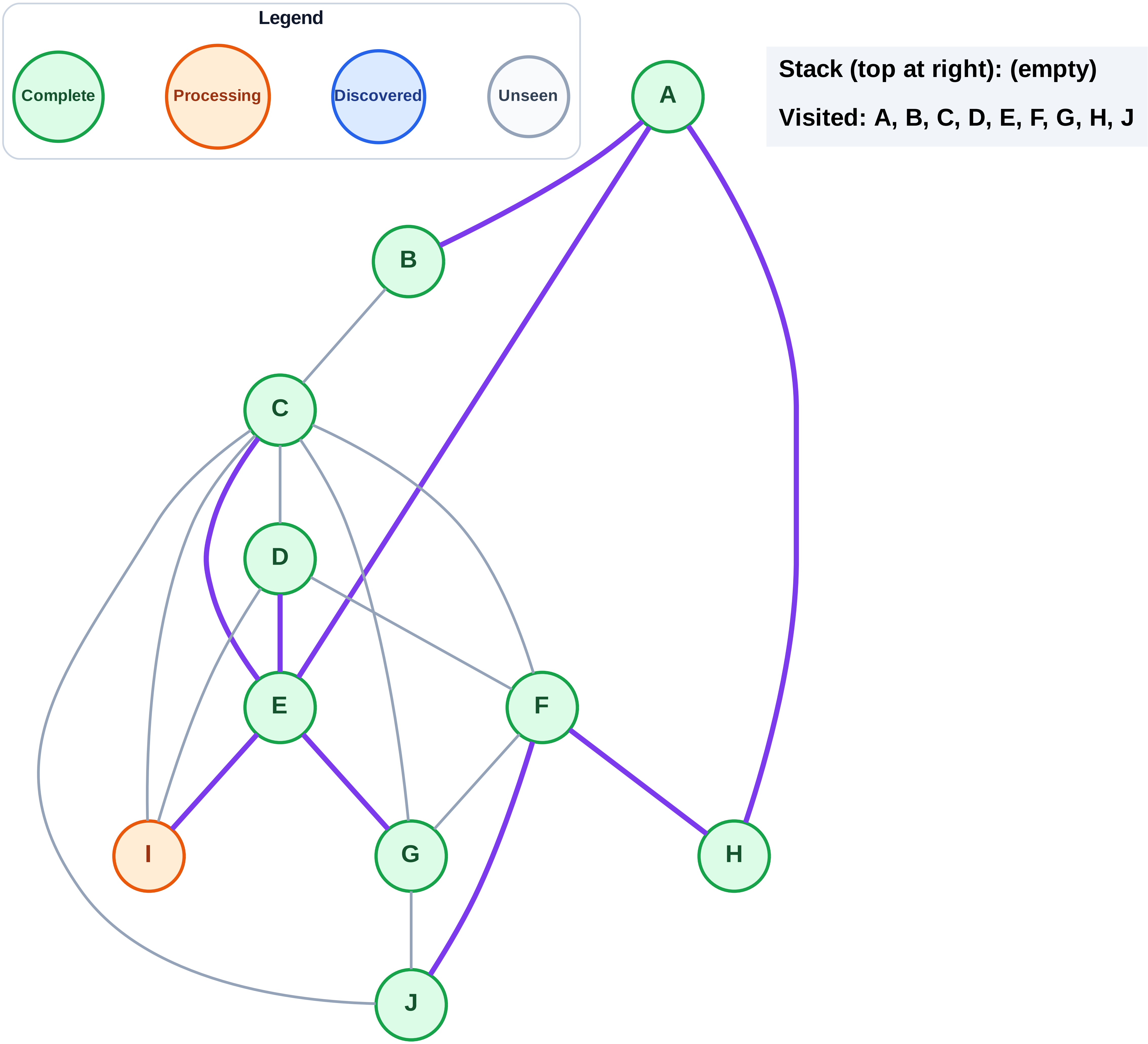
Step 18: Process node G




DFS Traversal
Step 19: No new neighbors from G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): I
Visited: A, B, C, D, E, F, G, H, J
A
B
C
D
E
F
I
G
H
J

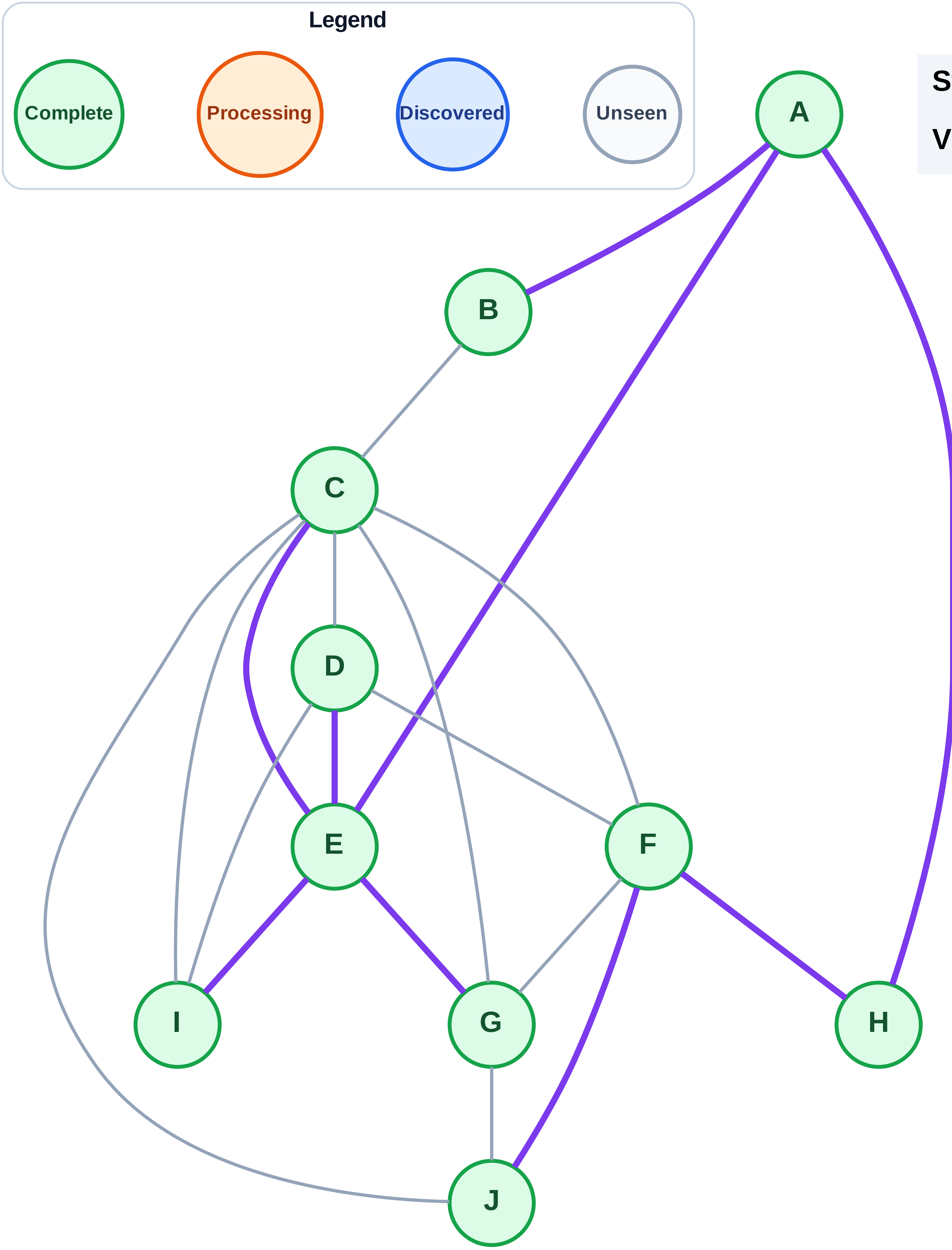
DFS Traversal

Step 20: Process node I



DFS Traversal

Step 21: No new neighbors from I



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J